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Adaptive voting stacking with attention mechanism for enhanced credit scoring☆

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ABSTRACT

Credit scoring is a cornerstone of modern financial decision making, yet it faces unique challenges due to the highly imbalanced nature of credit datasets and regulatory demands for interpretability and fairness. To overcome the generalization limitations of traditional predictive models under imbalanced distributions and noisy financial conditions, we propose adaptive attention-enhanced sequential stacking (AAESS). The proposed framework first integrates Pareto-improved feature selection with attention mechanisms to identify key variables more precisely. Then, it leverages robust statistical techniques for outlier detection to adaptively select optimal base learners for diverse feature distributions. Finally, it functions as a dynamic forecast combination strategy, using adaptive voting and weight allocation to enhance overall robustness. An embedded feature tracking mechanism further improves the interpretability and fairness of the model, offering transparent explanations that support the institutional need for model auditability. Experimental results on four real-world and highly imbalanced credit datasets demonstrate that AAESS achieves significant forecasting gains, with AUC improvements ranging from 4.7% to 9.3% and F1-score increases of over 6%, maintaining strong performance even under extreme class imbalance. The code is available at https://github.com/SHIMMER-0039/credit-scoring.git.

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1. Introduction

Credit scoring underpins virtually all lending, leasing, and buy-now-pay-later decisions in contemporary finance (Liu et al., 2023). Early practice centered on the qualitative Five Cs framework—character, capacity, capital, conditions, and collateral—whose subjectivity limited both scalability and accuracy. Over the past two decades, machine-learning techniques have progressively supplanted purely statistical approaches: interpretable models such as logistic regression (LR) and linear discriminant analysis (LDA) remain popular, yet they falter when confronted with the non-linearity and heterogeneity of modern credit data. This has motivated the adoption of more expressive learners—including decision trees (DT), random forests (RF), neural networks (NN) (West, 2000), support vector machines (SVM) (Danenas & Garsva, 2015), and gradient-boosted decision trees (GBDT) (Okesola et al., 2017)—which demonstrably lift predictive performance.

Despite these advances, recent regulatory developments have reframed the modeling agenda. The European Union's Artificial Intelligence Act (2024) requires credit algorithms to be traceable and explainable, while the U.S. Consumer Finance Protection Bureau's 2025 fair-lending

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examinations highlighted group-level fairness. The Basel Committee on Banking Supervision standard number 239 calls for robust data aggregation and verifiable risk reporting, obligations that weigh particularly heavily on small and mid-sized institutions. While ensuring strict legal compliance is primarily a multi-faceted institutional challenge, predictive models can support these regulatory goals by offering greater auditability. However, developing such transparent models is complicated by two endemic properties of credit datasets: extreme class imbalance (defaults are typically orders of magnitude rarer than non-defaults); and low-dimensional, noisy feature spaces, especially in niches such as micro-lending and consumer credit. Together, these factors expose the limitations of conventional and even state-of-the-art predictive models, signaling the need for architectures that can simultaneously address imbalance, noise, and interpretability, thereby aligning with the broader demand for algorithmic transparency (Sun et al., 2022).

The No Free Lunch theorem posits that no single classifier can dominate across all problem domains (Wolpert & Macready, 1997). Given the heterogeneity of financial datasets, this result sharply limits the effectiveness of standalone models in credit scoring. Ensemble learning therefore offers a more powerful alternative: by aggregating multiple base learners, an ensemble harnesses algorithmic complementarities, improving generalization under diverse data distributions and raising both accuracy and robustness—especially in small-sample, imbalanced, and high-noise settings (Rao et al., 2023; Shen et al., 2022). The attendant increase in architectural complexity, however, often deepens the model's opacity, reinforcing the black box perception that impedes regulatory compliance and erodes user trust. To reconcile performance with transparency, we incorporate a Pareto-guided feature selection stage. Grounded in the well-known Pareto principle in finance, this multi-objective filter makes variable contributions explicit, substantially enhancing the interpretability of the ensemble without compromising predictive power.

Following Pareto-based feature pruning, we quantify the marginal contribution of each retained variable to the discrimination of default cases, and rescale the predictors into proportional feature weights, thereby prioritizing the attributes that are most informative for the minority (default) class. Financial datasets are frequently characterized by idiosyncratic outliers—stemming from anomalous transactions, reporting errors, or macroeconomic shocks—which can compromise model training. To address this, the reweighted feature matrix is processed through a dedicated outlier detection module that integrates univariate techniques (interquartile range and Z-score tests) with multivariate approaches (isolation forest and local outlier factor). Should more than 5% of observations be identified as extreme, a robust Bayesian ridge estimator assigns instance-specific weights; otherwise, a Huber estimator mitigates the impact of moderate anomalies.

These sample weights inform an adaptive stacking ensemble comprising four heterogeneous base learners: random forests, XGBoost, AdaBoost, and LightGBM. Drawing inspiration from the attention mechanism in neural networks, the first-stage outputs of these learners are aggregated via a dynamic forecast combination strategy that dynamically amplifies the contributions of learners demonstrating superior recall for the minority class while attenuating those with weaker performance. In credit-scoring tasks marked by sample heterogeneity and severe class imbalance, even models with modest global performance may exhibit localized strengths in specific data subspaces. Aggregating their complementary perspectives, rather than relying on a single optimal predictive model, enhances generalization and stabilizes default probability estimations. Finally, an adaptive meta-learner—Bayesian ridge regression when outliers are sparse, or Huber regression when they exceed 5%—refines the aggregated scores, anchoring the final output in robust statistics to ensure resilience against financial noise while preserving interpretability.

This methodological framework systematically addresses outliers, class imbalance, and distributional shifts, yielding a credit risk model that achieves high discriminative power while maintaining transparency and interpretability through its explicit variable and instance weighting structure.

The remainder of this paper is organized as follows: Section 2 reviews related work on feature processing and ensemble models in credit scoring. Section 3 presents the AAESS model, including Pareto optimization and adaptive stacking. Section 4 describes the experimental setup across four credit scoring datasets. Section 5 analyzes the experimental results, comparing feature selection, model performance, and stacking approaches. Finally, concludes with a discussion of model interpretability and future research prospects.

2. Related work

2.1. Feature selection in credit scoring

Feature selection, or attribute selection, is essentially a preprocessing step where the algorithm identifies both important and irrelevant attributes from the entire set of available attributes (Nugroho & Setiawan, 2016). Feature selection is a pivotal step in credit scoring, as it identifies the most predictive features, eliminates redundancy, reduces model complexity, mitigates overfitting, and enhances generalization. Certain methods also transform or generate features to improve interpretability and capture complex relationships in credit data. For instance, synthesized features can better reflect customer credit behavior, while robust feature selection fosters model innovation and stability (Cui et al., 2021; Qummar et al., 2019; Tsai et al., 2021; Xiao et al., 2020).

Several studies have explored feature selection in stacking models for credit evaluations. Wang and Chi (2024) integrated cost-sensitive learning with stacking, using feature selection to identify an optimal subset for corporate bankruptcy prediction, pinpointing four key features critical to credit assessment. Wang et al. (2018) employed diverse training subsets, feature sampling, and parameter perturbations to ensure learner diversity, using logistic regression as a secondary learner to analyze XGBoost

outputs as an effective evaluation model. Trustorff et al. (2011) combined oversampling with Gaussian SMOTE and a genetic algorithm in their ESCRPGSE model to address class imbalance, selecting features via 10-fold cross-validation. Muslim et al. (2023) applied GLFS to a peer-to-peer (P2P) lending dataset, leveraging group lasso with the L2 norm to select or exclude feature groups. This was followed by random oversampling and stacking with LightGBM, XGBoost, and random forests, and logistic regression as the final estimator, significantly improving defaulter detection. Nalić et al. (2020) proposed an 'if_any' voting method to select optimal feature selection algorithms, introducing a local binary social spider algorithm with oppositional-based learning and an improved local search to address initialization and local optima issues. Zhou et al. (2022) extended single-feature selection to feature combination selection using a Fisher discriminant analysis F-score, optimizing combinations for maximum discriminative power in default prediction.

While the aforementioned approaches have advanced feature selection in credit scoring, they typically suffer from an inherent limitation: a tendency to optimize for a single objective, most often classification accuracy. Such a narrow focus overlooks the inherently multi-faceted nature of credit risk assessment, where tradeoffs among predictive performance, interpretability, and robustness are crucial for regulatory compliance and practical adoption. Moreover, existing strategies that attempt to combine multiple selection methods frequently lack a rigorous theoretical foundation to reconcile conflicts among competing feature rankings, thereby undermining consistency and reliability.

To address these gaps, our AAESS framework introduces a Pareto-optimal feature selection mechanism. Instead of maximizing a single performance criterion, this approach identifies a set of non-dominated solutions across a multi-dimensional objective space (e.g. AUC, recall, and model complexity). This ensures that the chosen feature subsets are not only predictive but also balanced and robust, directly aligning with the multi-objective requirements of real-world financial risk management.

2.2. Ensemble models

Ensemble models have become a cornerstone for addressing the complexities associated with credit scoring, especially in situations involving high-dimensional, noisy, or imbalanced datasets. These models leverage multiple base learners to enhance the robustness and accuracy of the prediction process. Ensemble methods can be broadly categorized into two types: homogeneous ensembles, which use multiple copies of the same base model trained on different data subsets or with varied parameters; and heterogeneous ensembles, which combine different types of models, such as decision trees, logistic regression, and support vector machines.

One of the significant advances in ensemble learning for credit scoring has been the development of stacking models. Stacking allows for the sequential integration of multiple classifiers, enabling the model to iteratively refine its decision-making process through successive layers of learning (Wang et al., 2025). This approach enhances the model's ability to capture complex patterns by progressively combining features and predictions from base learners. Recent improvements to stacking models, such as dynamic weight allocation, have introduced methods where models that perform better in certain scenarios are given more weight, allowing the ensemble to adapt to changes in data distributions over time (Yao et al., 2022). These adaptations are crucial for maintaining model performance across different datasets and business scenarios, especially in financial services, where the regulatory landscape and economic conditions can change rapidly.

Recent advancements in ensemble learning techniques have provided notable improvements. For instance, Liu et al. (2023) introduced a heterogeneous forest method to develop a flexible and interpretable cascade framework that improves the adaptability of tree-based models. Other studies have focused on combining feature selection and stacking strategies to enhance the model's generalization ability and reduce overfitting, ensuring more reliable credit predictions. Ala'raj and Abbod (2016) developed the ConsA rule, a classifier combination technique that uses adjacency graphs and multivariate adaptive regression splines to address complex, nonlinear relationships in credit data. De Bock et al. (2020) applied NSGA-II to optimize ensemble classifiers under uncertain misclassification costs, introducing a Pareto-optimal framework that balances multiple conflicting objectives in credit risk prediction. Xia et al. (2020) proposed a dynamic weight allocation strategy for tree-based ensemble models, enabling the system to better handle overfitting and adapt to local sample characteristics. Xiao et al. (2016) created a benchmark comparison framework for ensemble models, applying statistical analysis to identify the optimal model combinations for credit scoring. Finally, Zhang et al. (2021) introduced an anomaly detection voting mechanism combined with bagging-based balanced sampling to handle imbalanced credit datasets and improve classification accuracy.

Credit-scoring ensembles have pushed predictive accuracy forward, yet they still suffer from several systemic drawbacks: their multi-layer architectures frequently become so complex that they overfit imbalanced datasets, consume prohibitive computation in large-scale or real-time settings, remain sensitive to outliers, and fail to adapt when feature distributions shift across portfolios. Moreover, conventional feature selection pipelines often retain redundant or noisy variables, diluting model interpretability and weakening generalization—an unacceptable compromise in a heavily regulated, transparency-driven domain.

In particular, existing ensemble models face three systemic challenges that are especially salient in credit scoring. First, their architectural complexity often undermines interpretability, producing black box models that regulators and practitioners struggle to trust. Second, they are highly sensitive to outliers and noisy financial records, where even a few anomalous points can destabilize performance. Third, most weighting schemes are static and fail to adapt to the heterogeneous predictive strengths of base learners across different applicant sub-populations.

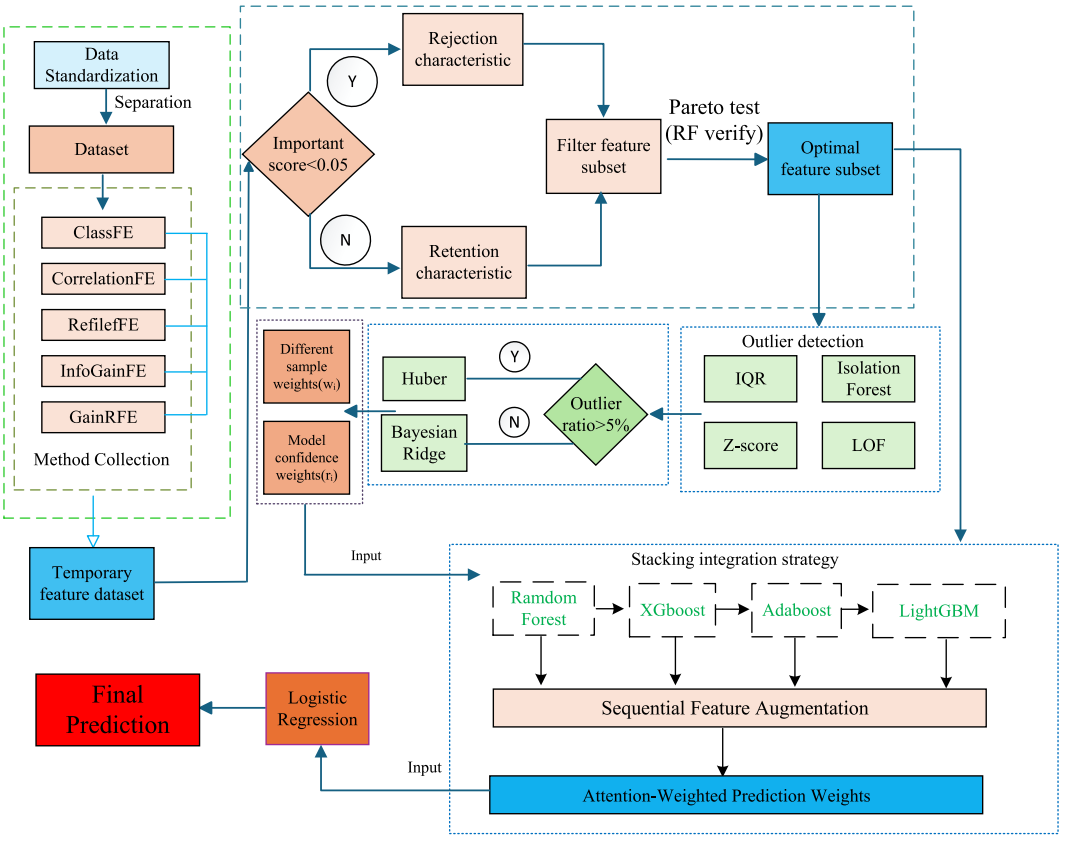


Fig. 1. Flowchart illustrating the process.

To remedy these issues, we propose the adaptive attention-enhanced sequential stacking (AAESS) framework. AAESS directly addresses the above limitations: it mitigates the outlier problem through a robust, adaptive weighting module based on Huber regression and Bayesian ridge regression, while its attention mechanism provides dynamic, performance-aware weighting to the base learners. By making this weighting process transparent, AAESS not only enhances robustness and predictive accuracy but also bridges the gap between performance and interpretability—an essential balance for modern financial risk management. Coupled with Pareto-optimal feature selection, which simultaneously optimizes accuracy, recall, and the area under the receiver operating characteristic (AUC) to retain only the most informative predictors, AAESS delivers a leaner and more interpretable ensemble that curbs overfitting, curtails computation, and maintains robustness across heterogeneous credit datasets—thereby addressing the chief limitations of existing stacking models.

3. Proposed methodology and the model

Our methodology is structured into three principal stages. First, we perform Pareto-optimal feature selection to identify the most informative predictors by jointly optimizing multiple performance metrics that are critical in credit risk management. Second, we introduce a robust weighting module that leverages outlier detection to mitigate the adverse impact of noisy or anomalous financial records, thereby enhancing model stability. Third, these components are integrated within a heterogeneous stacking ensemble, where an attention mechanism adaptively assigns weights to the outputs of diverse base learners, culminating in a unified and reliable prediction. The overarching objective is to achieve superior predictive performance while preserving a high level of interpretability. A detailed flowchart of the complete AAESS framework is provided in Fig. 1.

Rather than an arbitrary assembly of machine learning algorithms, the AAESS framework is theoretically grounded in the principles of dynamic forecast combination and robust statistics. Traditional ensemble methods in credit scoring often rely on static weighting. In contrast, our attention mechanism functions as a dynamic forecast combination strategy, where the influence of diverse base learners is continuously calibrated based on their localized predictive confidence across different borrower profiles. Furthermore, the integration of Huber regression and Bayesian ridge regression is driven by the inherent non-stationarity and idiosyncratic noise of financial datasets. By incorporating these robust statistical techniques, the framework effectively penalizes observational anomalies and quantifies epistemic uncertainty, ensuring that the dynamic forecast combination is anchored on theoretically sound parameter estimates.

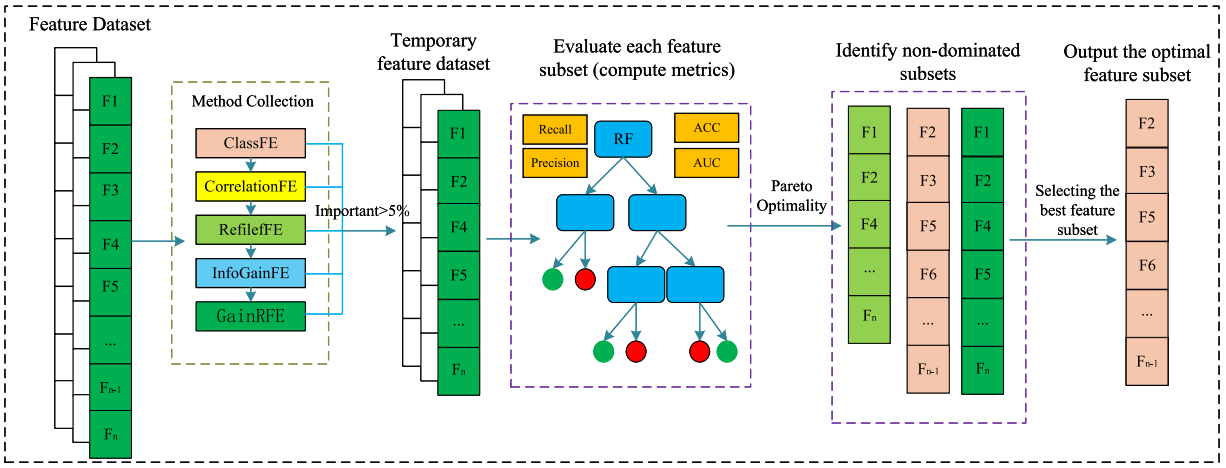


Fig. 2. Pareto feature selection.

3.1. Pareto-optimal feature selection and feature selection procedure

In the preprocessing stage, we employ Pareto optimality to enhance feature engineering for credit scoring, where balancing metrics like the AUC, recall, and interpretability are critical for regulatory compliance. Unlike traditional methods that optimize a single metric, risking overfitting and reduced robustness, Pareto optimization ensures simultaneous consideration of multiple metrics, yielding a balanced and robust model aligned with financial-sector requirements (see Fig. 2).

Originally derived from welfare economics, Pareto optimality describes a state of resource allocation where no individual's utility can be improved without reducing the utility of another.

In the classical economic framework, for an individual i, utility is defined as a function of resources:

$$U_i = f(x_{i1}, x_{i2}, \ldots, x_{in}), \tag{1}$$

where $x_{i1}, x_{i2}, \ldots, x_{in}$ represent allocated resources. For two individuals A and B with utilities $U_A = f(x_1, y_1)$ and $U_B = g(x_2, y_2)$, subject to total resource constraints

$$x_1 + x_2 = X, \quad y_1 + y_2 = Y, \tag{2}$$

a Pareto-optimal state is achieved when their marginal rates of substitution are equal:

$$\frac{\partial U_A/\partial x_1}{\partial U_A/\partial y_1} = \frac{\partial U_B/\partial x_2}{\partial U_B/\partial y_2}. \tag{3}$$

This condition ensures that no reallocation of resources can improve one utility without diminishing the other.

In the specific context of this paper, we adapt this theoretical framework to the domain of multi-objective feature selection. Here, 'utility' is no longer an abstract welfare function derived from resource consumption, but instead denotes the predictive performance of a feature subset evaluated under multiple classification metrics, such as accuracy, the AUC, recall, precision, and the F1-score. These metrics are treated as the dimensions of utility to be jointly optimized. Consequently, a Pareto improvement in our model is defined as a state where a candidate feature subset achieves a gain in at least one metric without strictly deteriorating the others:

$$\Delta U_A > 0, \quad \Delta U_B \geq 0, \tag{4}$$

indicating the potential for optimization if the system has not yet reached the Pareto frontier.

To enhance model robustness and reduce redundancy, we adopt a multi-method feature selection strategy combined with Pareto optimization. The overall procedure is summarized as follows.

First, the original feature set is evaluated independently by five different selection methods: classifier-based feature evaluation (ClassFE), correlation-based feature evaluation (CorrelationFE), information gain (InfoGainFE), relief-based feature evaluation (ReliefFE), and recursive feature elimination with gain (GainRFE). Each method computes an importance score for every feature according to its own criterion. Within each method, features with an importance score below the threshold of 0.05 are removed, yielding five candidate feature subsets.

Second, the candidate subsets are assessed using a random forest classifier under multiple performance metrics, including accuracy, AUC, precision, recall, and F1-score. These results are then compared in a multi-dimensional metric space. And then, to integrate the outcomes of different methods, we employ Pareto efficiency as the selection principle. Specifically, subsets that are dominated across all metrics are discarded, while those lying on the Pareto front are retained as efficient solutions. For features showing mixed importance, the Pareto test determines whether their removal improves at least one metric without deteriorating the others. Finally, among the Pareto-efficient subsets, the one that achieves the best tradeoff—particularly in terms of the AUC and recall, which are the most critical in credit scoring—is chosen as the final feature set for model training. The results of this process are also reflected in the feature selection tables presented in the manuscript.

Algorithm 1 Feature selection process for AAESS.

Input: Feature matrix $X \in \mathbb{R}^{n \times d}$, target vector y
Methods: ClassFE, CorrelationFE, GainRFE, InfoGainFE, and ReliefFE
Initialize feature selection methods: $\mathcal{M} = \{\text{ClassFE, CorrelationFE, GainRFE, InfoGainFE, ReliefFE}\}$
Initialize selected feature set: $\hat{X} = X$
for each method $m \in \mathcal{M}$ **do**
    Compute feature importance scores $S_m = m(X, y)$
    Filter step: Remove features with $S_m < 0.05$, updating $\hat{X}$
    Pareto evaluation: Apply Pareto test to assess the selected features in $\hat{X}$
    if Pareto test indicates improvement **then**
        Update $\hat{X}$ with features satisfying Pareto optimality
    else
        Retain current $\hat{X}$
    end if
end for
Final refinement: Apply ReliefFE to $\hat{X}$ for final feature optimization
Output: Optimized feature set $\hat{X}$

3.2. Outlier handling options

3.2.1. Outlier detection

To detect anomalous observations, we employ four complementary methods: the interquartile range (IQR), Z-score analysis, isolation forest, and the local outlier factor (LOF). The aggregated results of these detectors are then used to guide the choice of the weighting strategy. In particular, the 5% threshold for switching between Huber regression and Bayesian ridge regression (BRR) represents a critical parameter for adapting to data contamination. This choice is grounded in established principles of robust statistics and widely adopted practices in financial econometrics, where a 5% cutoff is commonly regarded as a reliable benchmark for identifying influential outliers. A comprehensive theoretical and empirical justification of this detection strategy and threshold setting is provided in Appendix A.

3.2.2. Robust attention-weighted mechanism for enhanced credit scoring

Credit scoring models face unique challenges due to the nature of financial data, which often include noise from data entry errors, imbalances from rare default events, and outliers from extreme credit histories. Traditional models may struggle with these issues, leading to reduced accuracy and reliability. The AAESS model's robust attention-weighted module addresses these challenges by integrating BRR and Huber Regression. These techniques provide a dual mechanism to handle noise and outliers effectively, ensuring that the model remains robust and interpretable even in the presence of imperfect data.

The AAESS model's robust attention-weighted module introduces a novel mechanism to improve model robustness and interpretability by integrating two complementary regression techniques: BRR and Huber regression. These methods work together to generate two key sets of information—sample weights w_i and model-level confidence γ_k—which are dynamically combined to refine predictions in the attention layer. Importantly, their roles are stage-specific: Huber regression is employed only during training when labels are available to compute residuals, while BRR operates in both training and inference. This distinction ensures that the framework is not only robust at learning but also feasible at deployment.

Sample weights w_i reflect the reliability of individual data points, allowing the model to mitigate the impact of outliers or uncertain observations. The two regression methods contribute to w_i in distinct yet complementary ways:

Huber regression (training stage only): When the detected outlier ratio exceeds 5%, Huber regression is activated during training. It adjusts the influence of each sample based on its residual $r_i = y_i - \hat{y}_i$:

$$w_i = \begin{cases} 1, & |r_i| \le \delta, \\ \dfrac{\delta}{|r_i|}, & |r_i| > \delta. \end{cases}$$

This approach reduces the weight of outliers, dampening their impact on parameter estimates and ensuring a more stable fit. Since this method requires true labels, it is not applied during inference.

Bayesian ridge regression (BRR): In environments characterized by prevalent anomalies, BRR leverages Bayesian principles to quantify epistemic uncertainty. For each sample i, the model computes a scalar predictive variance based on the feature space:

$$\sigma_i^2 = x_i^\top \Sigma x_i + \beta^{-1}, \tag{5}$$

where $\Sigma = (\alpha I + \beta X^\top X)^{-1}$ denotes the posterior covariance matrix of the weights, and β^{-1} represents the variance of the observation noise.

To incorporate this uncertainty into the attention mechanism, we construct a reliability-weighted feature vector $w_i \in \mathbb{R}^d$ by scaling the original feature vector inversely with the predicted variance:

$$w_i = \frac{1}{\sigma_i^2} \cdot x_i. \tag{6}$$

This formulation does not arise directly from BRR itself. BRR provides only the scalar variance σ_i^2; the vector w_i is an AAESS design choice that propagates sample-level uncertainty into feature-level attention. Samples with large predictive variance (σ_i^2) generate smaller w_i, diminishing their presence in the feature space. Importantly, since BRR relies solely on input features, this reliability estimate is fully label-free and usable during inference, enabling the model to down-weight uncertain samples without requiring residuals. The model-level confidence vector γ_k quantifies the reliability of each base learner. To support fine-grained attention, we define γ_k as a feature-wise confidence vector rather than a scalar.

When using BRR, $\gamma_k \in \mathbb{R}^d$ is derived from the diagonal of the posterior covariance matrix Σ_k, representing the precision of the learned coefficients:

$$\gamma_k = \left(\text{diag}(\Sigma_k)\right)^{-1}. \tag{7}$$

Each entry of γ_k corresponds to the inverse variance of the parameter for a specific feature, indicating how confidently the base learner estimates that feature. The dot product $\gamma_k^\top w_i$ in the attention layer thus captures the alignment between (i) reliable sample features and (ii) confident model parameters, forming a unified reliability-aware weighting signal.

When using Huber regression (training stage only): Huber regression does not naturally produce a Bayesian posterior covariance, but we can compute a robust covariance matrix $\hat{\Sigma}_k$ for its parameter estimates (e.g. via sandwich estimators in robust linear modeling). We then define

$$\gamma_k \propto \left(\text{trace}(\hat{\Sigma}_k)\right)^{-1}, \tag{8}$$

Since this definition yields a scalar, we broadcast the value across all d dimensions so that γ_k is dimensionally consistent with the BRR-based confidence vector. A smaller trace($\hat{\Sigma}_k$) implies less parameter uncertainty and hence higher model confidence. As this covariance relies on residual-based fitting, it is computed only during training and held fixed during inference.

Finally, the attention layer synthesizes these insights to produce the attention-weighted prediction inputs for each sample i and base learner k:

$$\tilde{y}_i^{(k)} = \alpha_{k,i} \cdot \hat{y}_i^{(k)}, \tag{9}$$

where $\hat{y}_i^{(k)}$ is the default probability predicted by the kth base learner, and $\alpha_{k,i}$ represents the normalized attention weight derived from γ_k and w_i (as detailed in Eq. (12)). This weighting mechanism operates on two levels: the sample dimension, where w_i reduces the influence of noisy or high-variance samples, effectively denoising the data; and the model dimension, where γ_k amplifies contributions from base learners that have demonstrated stable performance.

By explicitly integrating robust covariance matrices into the computation of model-level confidence under both BRR and Huber regression, the AAESS model maintains a consistent, rigorous framework for quantifying model reliability. The separation of training-time Huber regression and test-time BRR not only resolves the practical issue of label unavailability at inference, but also ensures that the weighting mechanism is robust, interpretable, and deployable in real credit scoring systems.

3.3. Adaptive attention-enhanced sequential stacking

3.3.1. Ensemble model selection

Ensemble learning improves classification accuracy by combining predictions from multiple base classifiers, with the results used to train a secondary classifier for the final prediction (Yin et al., 2023). Stacking models linearly combine the outputs of base classifiers, and the combined results are input into a logistic regression (LR) model. Common base models include random forests (RF) (Breiman, 2001), which reduces variance by aggregating predictions from decision trees; gradient boosting decision trees (GBDT), which iteratively refines model accuracy by correcting previous errors; and XGBoost (Chen & Guestrin, 2016), an enhanced version of GBDT that focuses on both accuracy and computational efficiency. LightGBM, another GBDT-based model, uses histogram-based discretization to reduce memory usage and computational complexity, while AdaBoost (Schapire, 2013) improves predictive power by iteratively refining weak classifiers. These methods, when combined in stacking, offer improved performance, robustness, and generalization across various datasets. In this study, we chose these models (RF, XGBoost, LightGBM, and AdaBoost) as base learners, due to their complementary strengths and their proven effectiveness at handling complex, high-dimensional, and imbalanced credit scoring tasks.

3.3.2. Adaptive attention-enhanced sequential stacking

Inspired by the attention mechanism in deep learning, we propose the adaptive attention-enhanced sequential stacking (AAESS) model, a hierarchical cascade architecture that processes information through multiple layers. Each layer comprises robust base learners, unlike traditional stacking methods that independently train models and concatenate their validation set predictions to form a secondary feature matrix. The primary goal of ensemble models is to effectively integrate base classifiers for superior performance. In this study, AAESS employs a stacking approach with two levels: the base level generates new features via diverse classifiers, and the meta-level constructs a meta-classifier using stacking to produce the final prediction.

AAESS utilizes an ensemble of classifiers $\{f_j\}_{j=1}^J$ ($J = 4$), denoted f_1 (random forests), f_2 (XGBoost), f_3 (LightGBM), and f_4 (AdaBoost). The meta-classifier g (logistic regression) integrates outputs for final predictions, selected for its computational efficiency and interpretability.

Using $K = 5$-fold cross-validation, the training set D_{train} is partitioned into $\{D_m^{(k)}\}_{m=1}^K$. Each classifier f_j trains on $D_{\text{train}} \setminus D_m^{(k)}$, predicting $\hat{y}_{j,i}^{(k)}$ on $D_m^{(k)}$. Meta-features are computed as

$$\hat{y}_{j,i} = \frac{1}{K}\sum_{k=1}^{K} \hat{y}_{j,i}^{(k)}, \tag{10}$$

where $\hat{y}_{j,i} \in [0, 1]$. AAESS sequentially augments features. For original features $X \in \mathbb{R}^{N\times M}$, the input to the jth classifier is

$$X^{(j)} = [X, \hat{Y}^{(1)}, \ldots, \hat{Y}^{(j-1)}], \tag{11}$$

where $\hat{Y}^{(l)} = [\hat{y}_{l,1}, \ldots, \hat{y}_{l,N}]^\top$, and $X^{(1)} = X$. Finally, the meta-classifier input vector Z is constructed by concatenating the original features with the attention-weighted prediction vectors:

$$Z = [X, \tilde{Y}^{(1)}, \tilde{Y}^{(2)}, \tilde{Y}^{(3)}, \tilde{Y}^{(4)}] \in \mathbb{R}^{N\times(M+J)}, \tag{12}$$

where $\tilde{Y}^{(j)}$ denotes the vector of attention-weighted outputs for the jth base learner.

3.3.3. Detailed description of the attention mechanism

To enhance model performance, particularly for minority classes (e.g. default samples in credit scoring), AAESS incorporates an attention mechanism that dynamically weights the predictions of base learners to adjust their contributions. The attention mechanism assigns weights $\alpha_{j,i}$ for each base learner j and sample i, forming the weight matrix $A = [\alpha_{j,i}] \in \mathbb{R}^{J\times N}$. The weights are calculated as

$$\alpha_{j,i} = \frac{\exp(\gamma_j^\top w_i)}{\sum_{l=1}^{J} \exp(\gamma_l^\top w_i)}, \tag{13}$$

where $\gamma_j \in \mathbb{R}^d$ is the confidence vector for base learner j, reflecting its reliability on the training data, and $w_i \in \mathbb{R}^d$ is the reliability vector for sample i, indicating its trustworthiness during training. These vectors are derived from parameters $\Theta = \{\theta_j\}_{j=1}^{J}$ of BRR or Huber regression, ensuring robustness to noise and outliers.

After retraining on D_{train}, test predictions $\{\hat{Y}_t^{(j)}\}_{j=1}^{J}$ are generated on $X_t \in \mathbb{R}^{N_t\times M}$. The meta-classifier input for sample i is constructed by concatenating the original features with the attention-weighted prediction vectors:

$$Z_{t,i} = [X_{t,i}, \alpha_{1,i}\hat{y}_{1,i}, \alpha_{2,i}\hat{y}_{2,i}, \alpha_{3,i}\hat{y}_{3,i}, \alpha_{4,i}\hat{y}_{4,i}], \tag{14}$$

producing the final prediction probability

$$p_i^{(\text{meta})} = g(Z_{t,i}; \beta), \tag{15}$$

where β denotes the meta-classifier parameters. The final label is as follows (Xia et al., 2018):

$$\hat{y}_i = \mathbb{I}(p_i^{(\text{meta})} \geq \mu), \quad \mu = 0.5. \tag{16}$$

This approach optimizes predictive accuracy through sequential feature augmentation and attention-based weighting. The whole process is shown in Fig. 3. The attention weights are obtained by combining sample-level weights w_i with model-level confidence γ_j, followed by a softmax normalization. This formulation makes the weighting transparent and reproducible, and it allows for reporting the contribution of each base learner and each sample explicitly.

4. Experimental setup

4.1. Credit assessment dataset

Shandong public funds. Released for a provincial big-data competition, this set contains 40,000 borrower records combining personal identifiers, housing-provident-fund deposits, and loan attributes. The task is to predict repayment defaults, supporting local risk-control initiatives and the development of Shandong's digital-economy infrastructure. Rich categorical fields and well-defined default labels make it a useful benchmark for consumer-credit modeling. The data are publicly available at http://data.sd.gov.cn/cmpt/cmptDetail.html?id=26.

Table 1
Summary of four credit scoring datasets.

Dataset	Samples	Features	Good/bad
Shandong	40,000	19	37,243/2757
Give	150,000	10	139,974/10,026
Fannie	369,335	18	349,458/28,877
BankFear	532,438	32	406,601/125,827

Fannie mae single-family. Curated by the Federal National Mortgage Association, this comprehensive mortgage repository provides quarterly 'acquisition' files (origination snapshots) and 'performance' files (monthly loan-level outcomes) from 2000–2020. Spanning roughly 20 million fixed-rate loans, it supports research on U.S. housing finance, prepayment, and default dynamics. The dataset's breadth and consistent formatting make it ideal for large-scale credit-risk benchmarking and stress-testing studies. It can be downloaded at https://capitalmarkets.fanniemae.com/credit-risk-transfer/single-family-credit-risk-transfer/.

Give me some credit. This is a popular Kaggle competition dataset of 150,000 U.S. consumers, containing demographic variables, revolving credit utilization, delinquencies, and debt ratios. Banks use similar attributes to estimate the probability of default for unsecured loans and credit-card portfolios. The clearly labeled targets and moderate dimensionality make it a standard test bed for feature engineering and imbalance-handling techniques. It is available on the Kaggle platform under 'Give Me Some Credit'.

BankFear. Compiled from Indian lending institutions that recently experienced surging non-performing assets, this set holds over 500,000 records with borrower profiles, loan terms, and repayment outcomes. Intended for a Kaggle challenge (bank-fears-loanliness), it asks participants to flag potential defaulters so banks can pre-emptively manage non-performing-asset exposure. The pronounced class imbalance and heterogeneous features provide a demanding benchmark for ensemble and anomaly-detection methods. The full files can be retrieved directly from Kaggle. Detailed information regarding the four credit scoring datasets is presented in Table 1.

4.2. Data evaluation models: Setup and evaluation

In this study, the dataset $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$ comprises feature sets x_i and credit outcomes $y_i \in \{0, 1\}$ (0 for non-default, 1 for default). It was partitioned into 70% training, 15% validation, and 15% test sets. The training set underwent five-fold cross-validation for hyperparameter tuning, with the validation set used for model selection and the test set for the final evaluation.

Feature selection employed five algorithms, validated by random forests, excluding features with importance scores below 0.05. This can speed up model training and cross-validation. A Pareto-based optimization then identified key features using multiple metrics. Outlier detection utilized four methods, triggering an adaptive weighting module: Huber regression for outlier rates above 5%, and otherwise Bayesian regression.

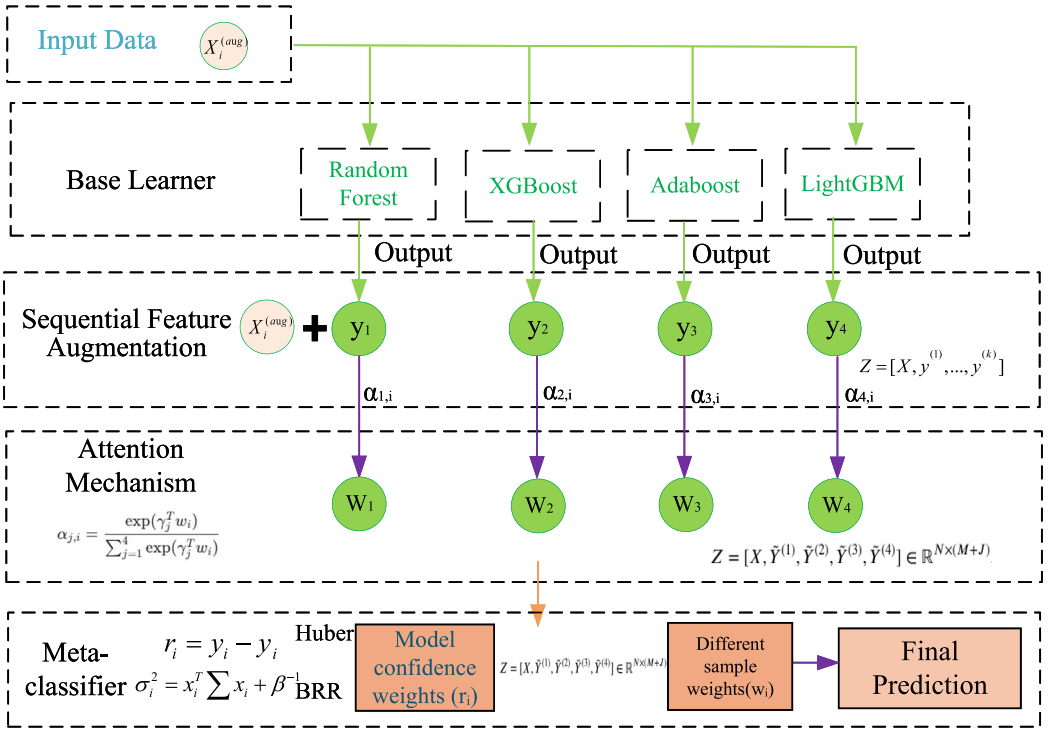


Fig. 3. AAESS process.

The AAESS model integrates four base classifiers: random forests (RF), XGBoost (XGB), LightGBM (LGB), and AdaBoost, with a logistic regression (LR) meta-classifier. BRR and Huber regression are used only as robust weighting modules for sample/learner weights. A self-attention mechanism weights base learners by cross-validated performance, sequentially combining outputs to enhance accuracy and efficiency.

This study conducts a systematic comparison of various credit scoring models, categorized into three groups: traditional statistical methods (LDA and LR), basic machine learning models (KNN, NN, and DT), and tree-based ensemble methods (RF, GBDT, XGBoost, and LightGBM).

Hyperparameter tuning optimized the number of estimators (trees for RF) for all base classifiers within [100, 1100] (step size: 200). For XGB and LGB, the maximum tree depth ranged from one to nine, and the learning rates were set in {0.01, 0.02, 0.05, 0.1, 0.2}. Optimal parameters were selected using a composite metric, equally weighting the AUC, Brier score (BS), H-measure (HM), and accuracy (Acc).

For clarity, the structure of the confusion matrix used in this study is summarized in Table 2, which distinguishes true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for credit scoring tasks.

Accuracy (Acc) measures the proportion of correctly classified instances:

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{17}$$

Table 2
Confusion matrix for credit scoring results.

Confusion matrix	Prediction	
Label	Bad (TP)	Good (TN)
Bad	TP	FN
Good	FP	TN

The area under the receiver operating characteristic curve (AUC) evaluates the overall ability of the model to distinguish between classes. It does not have a closed-form formula in this context.

Precision quantifies the proportion of true positive predictions among all positive predictions:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{18}$$

Recall (or sensitivity) measures the proportion of true positive predictions among all actual positive instances:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{19}$$

These are complemented by four additional metrics for a comprehensive credit scoring evaluation: the Brier score (BS), H-measure (HM), Type-I error (e-I), and Type-II error (e-II). The BS quantifies prediction errors:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2, \tag{20}$$

where p_i is the predicted probability and y_i the true label. HM addresses misclassification costs in imbalanced data

using a beta-distributed cost-weight function:

$$u(v) = \frac{v^{\alpha-1}(1-v)^{\beta-1}}{\int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,dt}, \quad \alpha < 1, \beta > 1, \tag{21}$$

with HM calculated as

$$\mathrm{HM} = 1 - \frac{\int_0^1 Q(T(v); b, u) u_{\alpha,\beta}(v)\,dv}{\pi_0 \int_0^1 u_{\alpha,\beta}(v)\,dv + \pi_1 \int_0^1 (1-v) u_{\alpha,\beta}(v)\,dv}, \tag{22}$$

where $Q(T(v); b, u)$ is the total loss, and π_0, π_1 are class priors. Type-I (e-I) and Type-II (e-II) errors assess false positive and false negative rates, respectively.

5. Experimental results

5.1. Ablation studies on feature selection and attention mechanism

To guarantee that the credit-scoring models were constructed on the most pertinent predictors, we implemented a multi-method feature selection strategy on both the Shandong and BankFear datasets. Table 3 presents the corresponding results. For each dataset, five distinct selection algorithms (denoted FS1–FS5) were applied; an 'X' indicates that the feature was retained by the given method, whereas the 'FIN' column reports the final subset forwarded to model training. This subset was determined on the basis of either cross-method consensus or a predefined acceptance rule.

In the Shandong dataset, all 10 candidate variables (F1–F10) satisfied the criterion and were therefore retained. By contrast, in the BankFear dataset, only seven variables (F1, F4, F7, F9, F10, F12, and F14) met the inclusion threshold, while the remaining attributes (F2, F3, F5, F6, F8, F11, F13, and F15–F18) were eliminated. This selective pruning underscores the effectiveness of the procedure at isolating the most informative predictors—particularly in the larger initial feature space of BankFear. By synthesizing several selection techniques, the resulting feature sets are robust and capture the key variables required for accurate credit-risk assessment.

To verify that the Pareto-optimal procedure genuinely improves feature quality for the downstream model, we conducted a dedicated ablation study. Random forests (RF) was selected as the validation backbone because its ensemble of decision trees produces stable estimates of feature importance while effectively capturing complex non-linear interactions, making it an appropriate benchmark for assessing the influence of the Pareto-derived feature subset. Model performance was evaluated with the four standard metrics widely employed in credit-risk modeling: accuracy, the AUC, recall, and precision. The comparative results after feature screening are reported in Table 4. For the same dataset, if the base classifier showed improved or equal performance after applying the Pareto-enhanced feature selection method, we highlight the corresponding metric values in **bold**.

On the more compact Give and Shandong datasets, the benefits are broader. Nearly all algorithms improve simultaneously in accuracy, AUC, recall, and precision, indicating that the Pareto-optimal subset removes noise without sacrificing discriminative power. Decision trees and AdaBoost, in particular, post substantial gains on Give, while random forests exhibits the largest jump on Shandong, underscoring that the proposed feature selection strategy is especially effective for weaker learners operating in limited-sample regimes. Collectively, the recall gains observed on all four datasets confirm that the selected attributes enhance the models' ability to identify positive (default) cases, validating the efficacy of the Pareto framework in elevating feature quality for credit-scoring tasks.

Table 3
Feature-selection results for the BankFear and Shandong datasets.[a]

Dataset	Feature	FS1	FS2	FS3	FS4	FS5	FIN
BankFear	F1	X		X		X	X
	F2		X		X		
	F3			X		X	
	F4						X
	F5	X					
	F6				X		
	F7	X			X		X
	F8		X			X	
	F9			X	X		X
	F10				X		X
	F11		X	X	X	X	
	F12	X		X			X
	F13		X		X		
	F14	X			X	X	X
	F15						
	F16						
	F17						
	F18						
Shandong	F1	X		X	X	X	X
	F2		X	X	X	X	X
	F3	X				X	X
	F4	X		X	X	X	X
	F5	X	X			X	X
	F6		X	X			X
	F7			X	X	X	X
	F8	X	X		X		X
	F9		X	X		X	X
	F10	X		X			X

[a] 'X' indicates that the feature is selected by the corresponding method; 'FIN' is the final retained subset.
F_i denotes the ith original predictor variable.
FS1–FS5 denote the five feature selection algorithms.

To evaluate the contribution of the attention mechanism in the AAESS model, we conducted an ablation study comparing the full AAESS model with a variant lacking the attention component, termed BaseStack. Table 5 presents the performance metrics for both models across four credit scoring datasets: Give, Shandong, Fannie, and Bankfear. The results demonstrate that AAESS consistently outperforms BaseStack in key metrics such as the accuracy, AUC, and Brier score across all datasets. For example, on the Bankfear dataset, AAESS achieves substantial improvements, with the accuracy increasing from 0.8276 to 0.9232 and the AUC from 0.8131 to 0.9473. However, BaseStack shows better performance in specific cases, such as the H-measure and Type-II errors for the Give and Fannie datasets, indicating tradeoffs in certain metrics. Overall, the attention mechanism significantly enhances AAESS's predictive accuracy and calibration, particularly for imbalanced and complex financial data, underscoring its value in credit scoring applications.

Table 4
Performance metrics of the features selected via Pareto filtering.

Dataset	Algorithm	Accuracy (↑)	AUC (↑)	Recall (↑)	Precision (↑)
Shandong	LR	**0.93575**	**0.82981**	**0.08614**	0.63880
	LDA	0.93300	**0.82830**	**0.11235**	0.49180
	DT	0.93675	0.75920	0.55430	0.52482
	KNN	0.93675	0.74531	0.21348	0.57000
	Adaboost	**0.95115**	**0.89905**	**0.33333**	**0.84761**
	RF	**0.96250**	**0.93496**	**0.46816**	**0.93984**
	GBDT	**0.95550**	**0.91962**	**0.38202**	**0.88695**
	XGBOOST	**0.96150**	**0.93326**	0.46441	**0.91851**
	LightGBM	**0.96375**	**0.93863**	**0.52434**	**0.88607**
Bankfear	LR	**0.77837**	**0.77085**	**0.25043**	**0.57511**
	LDA	**0.77422**	0.75904	0.26730	0.54922
	DT	**0.85613**	**0.80688**	**0.71321**	**0.69035**
	KNN	**0.77640**	0.69445	0.32100	0.54881
	Adaboost	**0.83047**	**0.80969**	0.30785	**0.84761**
	RF	**0.85863**	**0.89885**	**0.43838**	**0.93124**
	GBDT	**0.89560**	**0.90333**	**0.32615**	**0.92698**
	XGBOOST	**0.92023**	**0.93618**	**0.32615**	**0.94384**
	LightGBM	0.92203	**0.92618**	**0.60359**	**0.92286**
Fannie	LR	**0.92378**	**0.78751**	**0.02450**	**0.50364**
	LDA	0.92356	**0.79173**	0.02379	0.47517
	DT	**0.92405**	**0.79004**	0.01598	**0.56962**
	KNN	0.92394	**0.79674**	**0.00639**	**0.62068**
	Adaboost	**0.92324**	**0.81211**	**0.04865**	**0.46757**
	RF	0.92432	**0.81479**	0.01988	0.61538
	GBDT	**0.92459**	**0.83003**	**0.05788**	**0.55254**
	XGBOOST	**0.92464**	**0.83006**	**0.05823**	**0.55593**
	LightGBM	**0.92467**	**0.82813**	**0.04829**	**0.57142**
Give	LR	**0.93766**	**0.84509**	**0.15752**	0.59386
	LDA	**0.93786**	**0.85651**	**0.28353**	0.26702
	DT	**0.90020**	**0.62384**	**0.29878**	**0.26702**
	KNN	**0.93933**	**0.85064**	**0.16666**	**0.64566**
	Adaboost	**0.93888**	**0.86802**	**0.22560**	**0.58730**
	RF	**0.93993**	**0.87271**	0.17987	**0.65313**
	GBDT	**0.94033**	**0.87267**	**0.21951**	**0.62973**
	XGBOOST	**0.94013**	**0.87248**	**0.18292**	**0.65693**
	LightGBM	**0.93920**	**0.87286**	**0.20325**	**0.60975**

Table 5
Comparison of BaseStack and AAESS on four datasets.

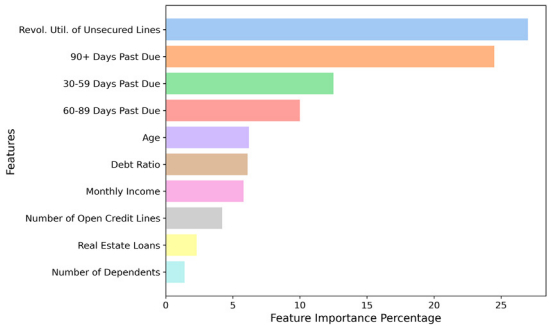
index	Give		Shandong		Fannie		Bankfear	
	BaseStack	AAESS	BaseStack	AAESS	BaseStack	AAESS	BaseStack	AAESS
Accuracy (↑)	0.9401	**0.9406**	0.9560	**0.9665**	0.9211	**0.9249**	0.8276	**0.9232**
AUC (↑)	0.8708	**0.8796**	0.9101	**0.9448**	0.8109	**0.8445**	0.8131	**0.9473**
Brier Score (↓)	0.0485	**0.0471**	0.0370	**0.0279**	0.0644	**0.0611**	0.1282	**0.0620**
H-measure (HM) (↑)	**0.3482**	0.2889	0.5686	**0.6762**	**0.1277**	0.0923	0.4775	**0.8262**
Type-I error (↓)	0.0103	**0.0061**	0.0067	**0.0029**	0.0083	**0.0007**	**0.0183**	0.0291
Type-II error (↓)	**0.7581**	0.7857	0.5655	**0.4292**	**0.9250**	0.9371	0.6679	**0.2304**

5.2. Feature importance analysis based on selection stage

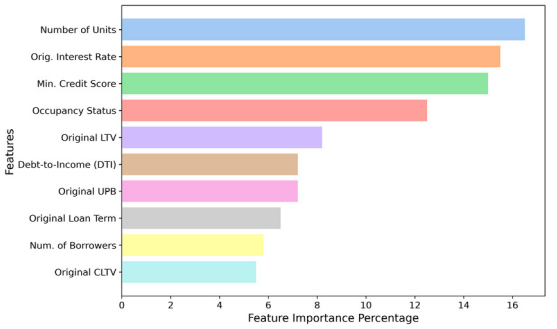
To analyze the intrinsic risk characteristics of the datasets identified during the Pareto-optimal feature selection stage, we examined the feature importance scores derived from the random forest verifier (as shown in Fig. 4). It is important to note that this analysis reflects the relevance of features prior to the stacking ensemble, providing insights into the dataset's risk drivers rather than the final model's decision weights.

For consumer-credit data (Fig. 4(a)) the two most influential variables are the revolving utilization of unsecured credit lines (27%) and the number of times 90+ days past due (23%), reflecting, respectively, current debt usage and severe delinquency history. In small-business loans (Fig. 4(c)) the total current balance and the number of open accounts dominate, signaling how overall leverage and portfolio complexity elevate default risk. Mortgage data (Fig. 4(b)) highlight property-specific factors—the number of property units, original interest rate, minimum credit score, and occupancy type—which jointly capture loan cost, borrower quality and collateral pressure. Finally, public-fund records (Fig. 4(d)) show that personal account status, employer economic type, industry sector and contribution base are decisive, linking repayment ability to job stability and income level.

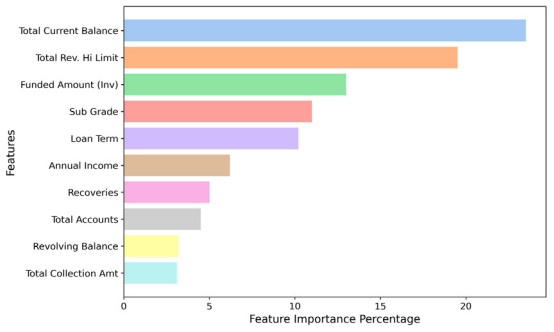
The subsequent AAESS framework builds upon these identified signals. By (i) down-weighting noisy samples via Bayesian ridge or Huber residuals, and (ii) up-weighting reliable base learners through posterior confidence, the model effectively leverages these key predictors (x_1,



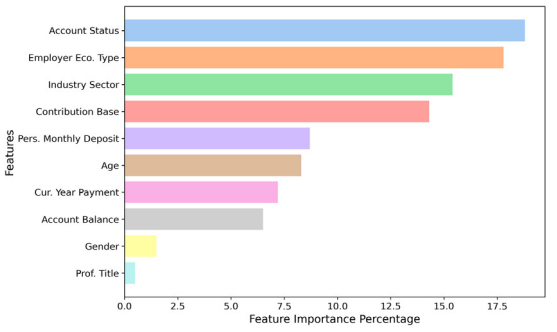
(a) Importance score for Give



(b) Importance score for Fannie



(c) Importance score for Bankfear



(d) Importance score for Shandong

Fig. 4. Feature importance scores derived from the random forest verifier during the selection stage.

x_7, x_{10}, etc.) to enhance robustness. The specific model-level interpretability and the contribution of base learners are discussed in Section 5.8.

5.3. Performance comparison of credit scoring models

5.3.1. Comparison with basic machine learning models

The figures above display the receiver operating characteristic (ROC) curves of different credit scoring models across four credit scoring datasets. Fig. 5(a) shows the ROC curve for the Bankfear dataset, Fig. 5(b) presents the ROC curve for the Fannie dataset, Fig. 5(c) depicts the ROC curve for the Give dataset, and Fig. 5(d) compares the ROC curves for the Shandong dataset. In Fig. 5(a), it is evident that the AAESS model achieves the largest AUC on the Bankfear dataset. The boosting ensemble models generally outperform simpler machine learning models. The DT model shows a smaller AUC area across all four datasets, highlighting its lower robustness. On both the Bankfear and Shandong datasets, the AUCs for DT and KNN are notably smaller compared to other models, indicating the poor adaptability of these models to the datasets.

In the Bankfear and Fannie datasets, ensemble models such as XGBoost and LightGBM also demonstrate high AUC values, closely matching those of the AAESS model and significantly surpassing RF, Adaboost, and GBDT. This indicates that models enhanced from GBDT, like XGBoost and LightGBM, are more suitable for credit scoring datasets, providing insight for selecting base learners in model construction.

In the Give dataset, models such as DT, LDA, and KNN show smaller AUC values compared to ensemble methods, further validating the superiority of ensemble algorithms. A detailed comparison reveals that AAESS, due to the incorporation of LightGBM and XGBoost, further enhances the overall performance of advanced ensemble algorithms in the Fannie and Give datasets.

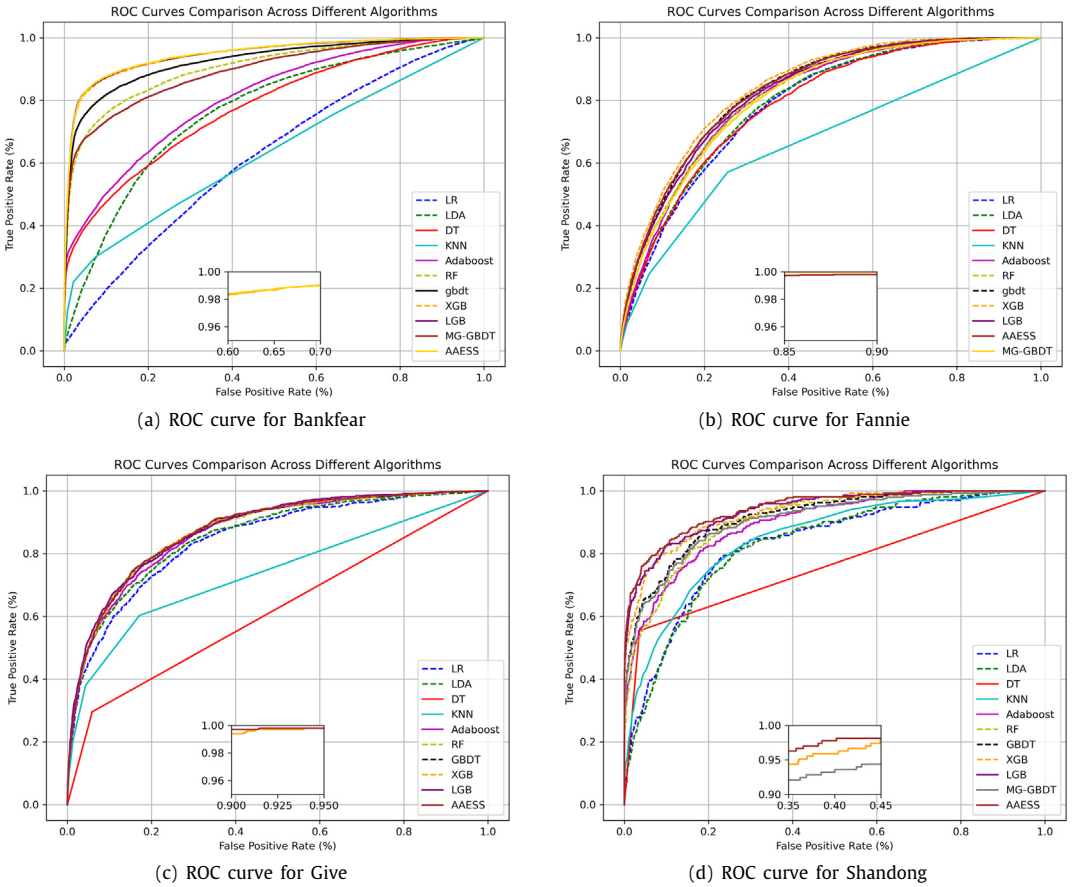
From the Tables 6–9, it can also be observed that the AUC scores for all credit scoring methods on the Shandong and Bankfear datasets are higher than the AUC scores on the Fannie and Give datasets. This is attributed to the inherent complexity of the credit scoring datasets. The performance of DT and KNN on the Shandong dataset is relatively poor. This is because DT typically uses criteria such as information gain and Gini index for node splitting. These criteria, when applied to imbalanced datasets, tend to favor features representing the majority class as the splitting points, leading to a bias towards majority class classification. Consequently, this results in a higher likelihood of samples being classified as the majority class.

KNN, on the other hand, determines the sample's class based on distance metrics. In imbalanced datasets, it is more likely to classify minority-class samples as the majority class, due to the proximity to the more frequent class, thus failing to accurately recognize minority-class samples. Since credit samples are generally more numerous than default samples, this phenomenon can occur, affecting the accuracy of the models in identifying minority class instances.

Table 6 summarizes the results on the Bankfear dataset. After the Pareto-based feature selection and adaptive

Table 6
Performance metrics for different algorithms on the Bankfear dataset.

Algorithm	Accuracy (↑)	AUC (↑)	BS (↓)	HM (↑)	Type-I error (↓)	Type-II error (↓)
LR	0.76620	0.62103	0.17618	0.34744	0.00324	0.97544
LDA	0.77614	0.76468	0.15263	0.36993	0.06873	0.72287
DT	0.82429	0.77807	0.13358	0.46552	0.01976	0.67733
KNN	0.77723	0.63363	0.18170	0.38933	0.05664	0.16612
Adaboost	0.82995	0.80837	0.23908	0.46590	**0.00708**	**0.16296**
RF	0.85440	0.90382	0.10199	0.57401	0.00859	0.58632
GBDT	0.89118	0.92588	0.08242	0.72044	0.01560	0.40868
XGBOOST	0.92023	0.94571	0.06415	0.81586	0.02535	0.25479
LightGBM	0.92117	0.94668	0.06326	0.81825	0.02508	0.25170
AAESS	**0.92319**	**0.94725**	**0.06198**	**0.82617**	0.02905	0.23039



(a) ROC curve for Bankfear

(b) ROC curve for Fannie

(c) ROC curve for Give

(d) ROC curve for Shandong

Fig. 5. ROC curve for different datasets.

weighting stages, AAESS attains the highest accuracy, AUC, Brier score (lowest), and H-measure, underscoring its capacity to translate feature-level gains into end-to-end risk reduction. Notably, while AdaBoost delivers competitive Type-I/Type-II error tradeoffs, and conventional learners such as LR and KNN suffer from pronounced AUC deficits, AAESS surpasses all bagging baselines and even the strongest boosting baselines (GBDT, XGBoost, and LightGBM). This confirms that the *sequential attention + Bayesian/Huber meta-learning design* offers a tangible advantage beyond what off-the-shelf boosting can achieve.

A similar pattern emerges in Table 7. On the much larger Fannie dataset, AAESS posts the best Brier score and the second-best accuracy and H-measure, while maintaining a Type-II error only marginally higher than the GBDT optimum. The consistent Brier-score superiority demonstrates that the proposed Pareto-optimal feature subset reduces probability calibration errors even at scale. Furthermore, the results show that models with fewer Type-I errors (e.g. LR and DT) do so at the expense of markedly higher Type-II errors—a cost imbalance that AAESS alleviates without sacrificing discrimination (AUC). Taken

Table 7
Performance metrics for different algorithms on the Fannie dataset.

Algorithm	Accuracy (↑)	AUC (↑)	BS (↓)	HM (↑)	Type-I error (↓)	Type-II error (↓)
LDA	0.92364	0.79425	0.06413	0.04665	0.00213	0.97549
LR	0.92381	0.78749	0.06455	0.01539	0.00058	0.99218
DT	0.92399	0.78932	0.06407	0.01750	0.00046	0.99112
KNN	0.91595	0.67061	0.07466	**0.13537**	0.01437	0.96966
Adaboost	0.92286	0.81346	0.23494	0.08715	0.00457	0.97256
RF	0.92435	0.81699	0.06210	0.03522	0.00084	0.98188
GBDT	0.92462	0.83194	0.06201	0.10482	0.00383	0.94211
XGBOOST	0.92229	0.84309	0.06308	0.07894	0.00249	0.95770
LightGBM	0.92185	0.82642	0.06308	0.02103	**0.00026**	0.98933
AAESS	**0.92493**	**0.84452**	**0.06107**	0.09226	0.00067	**0.93708**

Table 8
Performance metrics for different algorithms on the Give dataset.

Algorithm	Accuracy (↑)	AUC (↑)	BS (↓)	HM (↑)	Type-I error (↓)	Type-II error (↓)
LDA	0.93786	0.85651	0.05325	0.37449	0.01619	0.79645
LR	0.93786	0.84485	0.05023	0.24959	0.00734	0.84247
DT	0.89966	0.61834	0.10005	0.27189	0.05346	0.97834
KNN	0.93520	0.73722	0.05583	0.29257	0.01260	0.80341
Adaboost	0.93888	0.86802	0.23372	**0.32599**	0.01040	0.79037
RF	0.93993	0.87240	0.04754	0.28321	0.00676	0.81910
GBDT	0.93920	0.87339	0.04758	0.30909	0.00941	0.79268
XGBOOST	0.94013	0.87412	0.04731	0.31554	0.00806	0.80962
LightGBM	0.94020	0.87651	0.04725	0.30085	0.00614	0.80776
AAESS	**0.94061**	**0.87963**	**0.04705**	0.28890	**0.00607**	**0.78571**

Table 9
Performance metrics for different algorithms on the Shandong dataset.

Algorithm	Accuracy (↑)	AUC (↑)	BS (↓)	HM (↑)	Type-I error (↓)	Type-II error (↓)
LDA	0.93300	0.82830	0.05526	0.36364	0.00775	0.88764
LR	0.93325	0.61332	0.06408	0.34251	**0.00000**	0.91386
DT	0.93675	0.75920	0.06325	0.40474	0.03335	0.79452
KNN	0.92800	0.84829	0.05027	0.38721	0.00025	0.92134
Adaboost	0.95150	0.89905	0.23170	0.47849	0.00434	0.66662
RF	0.96250	0.93493	0.04252	0.35258	0.00051	0.78277
GBDT	0.95550	0.91692	0.03269	0.62829	0.00453	0.46923
XGBOOST	0.96375	0.93326	0.03215	0.61691	0.00275	0.43446
LightGBM	0.94020	0.93863	0.02950	0.65882	0.00452	0.46068
AAESS	**0.96650**	**0.94484**	**0.02788**	**0.67619**	0.00294	**0.42923**

together, the PDF evidence substantiates AAESS as a robust, scalable, and better-calibrated alternative to both classical and state-of-the-art ensemble credit scorers.

Table 8 shows the performance of various models on the small-scale credit scoring dataset, Give. Compared to the Bankfear and Fannie datasets, LR and LDA performed better on the smaller dataset, with Acc and AUC values surpassing those of KNN and comparable to those of Adaboost. Due to the lower complexity of the DT model, it exhibited the lowest AUC value. In terms of errors, although LightGBM did not achieve the best performance in terms of either Type-I or Type-II errors, its rankings were relatively high. The performances of XGBoost and GBDT were also commendable, further confirming the improvement in AUC values provided by tree-based machine learning methods. Additionally, AAESS ranked highly across multiple metrics, demonstrating its potential on small-scale datasets. The performance improvements of ensemble models across all metrics further validated the effectiveness of ensemble methods.

Table 9 shows the performance of various credit scoring models on the Bankfear dataset. The standalone LR model achieved a Type-I error rate of 0 but exhibited a very high Type-II error rate. This is due to the tendency of maximum likelihood estimation to assign higher optimization weights to correct classifications of the majority class, thereby neglecting correct classifications of the minority class. On this dataset, KNN outperformed its results on other datasets, possibly because the local structure of the samples was more apparent, with class labels being more consistent among neighboring sample points, which KNN effectively leveraged. Similar to conclusions drawn from other datasets, boosting ensemble models proved to be more effective for credit scoring than bagging-based ensemble models. On the Shandong dataset, the AAESS model achieved the best scores in metrics such as the Acc, AUC, BS, HM, and Type-II errors, indicating that AAESS is a good choice for reducing prediction errors and improving the scoring performance on imbalanced datasets.

5.3.2. Comparison between isomerism and heterogeneity

To validate the effectiveness of the heterogeneous ensemble framework, we compared the performance of AAESS with several deep cascade ensemble models, including homogeneous deep cascade ensemble models and heterogeneous deep ensemble methods. Additionally, to

Table 10
Comparison of performance metrics for homogeneous and heterogeneous stacking ensembles across different credit scoring datasets.

	Homogeneous stacking ensemble						Heterogeneous stacking ensemble			
Base learner	DT	LDA	LR	RF	XGB	LGB	AAESS-LR	ExTree-tree	AAESS-TREE	AAESS
Shandong										
Accuracy (↑)	0.95775	0.95175	0.95925	0.95675	0.96665	0.96655	0.96625	0.96475	0.96275	**0.96665**
AUC (↑)	0.91364	0.88010	0.90037	0.93641	0.93741	0.94286	0.94384	0.93756	0.93196	**0.94484**
BS (↓)	0.03603	0.07697	0.07198	0.03454	0.02866	0.02844	0.02796	0.03019	0.03141	**0.02788**
HM (↑)	0.57845	0.49907	0.55701	0.63584	0.65865	0.66789	0.67380	0.65693	0.64775	**0.67619**
e-I (↓)	0.00428	**0.00003**	0.00504	0.00107	0.00509	0.00375	0.00267	0.00241	0.00283	0.00294
e-II (↓)	0.57678	0.72285	0.63003	0.63296	0.43824	0.04644	0.46816	0.49443	0.43406	**0.42923**
Fannie										
Accuracy (↑)	0.92405	0.92424	0.92359	0.92318	0.92392	0.92454	0.92418	0.92470	0.92450	**0.92493**
AUC (↑)	0.79602	0.81332	0.80162	0.81727	0.83276	0.83317	0.83118	0.84114	0.83116	**0.84452**
BS (↓)	0.06348	0.06243	0.06384	0.06291	0.06194	0.06169	0.06170	0.06224	0.06161	**0.06107**
HM (↑)	0.25645	0.28137	0.25953	0.29096	0.32400	**0.32235**	0.10029	0.09804	0.09183	0.09226
e-I (↓)	0.00114	0.00023	0.00191	**0.00006**	0.00332	0.00399	0.00296	0.00508	0.00273	0.00067
e-II (↓)	0.98260	0.99707	0.97905	0.99600	0.94797	0.94141	0.94530	0.98290	0.95082	**0.93708**
Bankfear										
Accuracy (↑)	0.91413	0.91376	0.91378	0.83921	0.92202	0.92208	0.91679	0.87705	0.91702	**0.92319**
AUC (↑)	0.79116	0.81136	0.80864	0.88550	0.94615	0.94647	0.94051	0.91707	0.94186	**0.94725**
BS (↓)	0.06402	0.06292	0.06350	0.11154	0.06303	0.06260	0.06643	0.09300	0.06646	**0.06198**
HM (↑)	0.24228	0.27818	0.27337	0.49704	0.71342	0.71411	0.81120	0.74619	0.75803	**0.82617**
e-I (↓)	0.00108	**0.00003**	0.00029	0.00630	0.02565	0.02580	0.03252	0.01531	0.03033	0.02905
e-II (↓)	0.98189	0.99964	0.99929	0.65892	0.24632	0.24560	0.24623	0.46919	0.25233	**0.23039**
Give										
Accuracy (↑)	0.93520	0.93906	0.93866	0.93753	0.93780	0.93873	0.94020	0.93490	0.93980	**0.94061**
AUC (↑)	0.86713	0.85372	0.85597	0.84515	0.86487	0.87337	0.87370	0.87014	0.86690	**0.87963**
BS (↓)	0.04760	0.04880	0.04916	0.04927	0.04999	0.04884	0.04712	0.04748	0.04765	**0.04705**
HM (↑)	0.26451	0.28257	0.22689	0.19105	0.30216	**0.33646**	0.28752	0.26517	0.30162	0.28890
e-I (↓)	0.00656	0.00784	**0.00506**	0.00624	0.01077	0.01198	0.00613	0.00813	0.00627	0.00607
e-II (↓)	0.83333	0.81707	0.86280	0.83441	0.79471	0.76321	0.81605	0.83638	0.80182	**0.78571**

verify whether Bayesian weight allocation is an optimal allocation method, we used an LR-based method for weight allocation to check if the results were reasonable. Given the large-scale credit scoring in this study, we employed efficient machine learning-based individual classifiers (DT, LDA, and LR), as well as advanced ensemble methods (RF, XGBoost, and LightGBM), as the base learners for homogeneous ensemble learning. Correspondingly, for the heterogeneous deep cascade model, we adopted the ExTree-tree model (DT, RF, and Adaboost) and the AAESS-TREE model (DT, GBDT, and XGB). To assess the impact of Bayesian and Huber methods in noise detection, we conducted an ablation study by replacing them with LR for handling noisy samples, thereby evaluating the contribution of these advanced methods to the overall performance.

Table 10 showcases the performance of AAESS compared to various deep cascading ensemble models across four datasets: Give, Shandong, Bankfear, and Fannie. Measured by the area under the ROC curve (AUC) and Brier score (BS), AAESS consistently surpassed heterogeneous deep cascading models such as ExTree-tree and AAESS-TREE. For instance, on the Shandong dataset, AAESS recorded an AUC of 0.94484, outperforming ExTree-tree's 0.93756 and AAESS-TREE's 0.93196, alongside a BS of 0.02788, which is lower than 0.03019 and 0.03141 for the respective models. This pattern held across all datasets. Combined with findings from Tables 8–9, which evaluate classical credit scoring techniques (DT, LDA, and LR) and homogeneous ensemble methods (RF, XGBoost, and LightGBM), the adaptive deep structure of machine learning classifiers markedly improves credit scoring performance, affirming the cascading ensemble framework's strength. Further analysis of homogeneous and heterogeneous deep cascading models shows that integrating robust classifiers like RF, XGBoost, and LightGBM into the cascading structure delivers significant gains. Among heterogeneous models and those weighted by logistic regression (AAESS-LR), AAESS's adaptive weighting—utilizing Bayesian and Huber methods—secured the highest AUC and BS across all datasets, with standout accuracy scores on smaller datasets, such as 0.96665 for Give and 0.93708 for Fannie. AAESS achieves the lowest Type-II error (false negative rate) among all models across every dataset—a property of particular significance in credit assessment, where minimizing missed detections of risky applicants is critical for effective risk management and regulatory compliance.

The incorporation of Bayesian and Huber regression in AAESS significantly enhances the model's robustness to noise and outliers, offering superior stability over the simpler logistic regression (LR) approach, which lacks such adaptive features. In contrast, the LR method, while computationally efficient, relies on a linear assumption and fixed weighting, making it less capable of adapting to the complex, noisy nature of real-world credit data. This enhanced robustness translates into improved predictive accuracy, particularly for minority classes such as defaulting applicants, where noise and imbalance are prevalent. These findings underscore AAESS as an exceptional tool for probabilistic credit scoring predictions, adept at distinguishing between good and bad applicants with greater precision.

In order to evaluate the computational efficiency and practicality of the proposed framework, we compared

Table 11
Computational cost comparison of ensemble models for credit scoring (unit: seconds).

Phase/Dataset		DT	LDA	LR	RF	XGB	LGB	AAESS
Training	Shandong	0.974	0.079	0.073	5.012	0.538	1.508	8.207
	Give	0.412	0.043	0.031	1.911	0.490	1.684	4.581
	BankFear	1.534	0.296	0.275	11.589	3.049	2.818	19.626
	Fannie	15.067	0.811	2.743	37.343	14.557	5.254	75.847
Prediction	Shandong	0.006	0.002	0.003	0.109	0.005	0.105	0.230
	Give	0.001	0.001	0.001	0.073	0.004	0.095	0.176
	BankFear	0.015	0.008	0.012	0.471	0.026	0.275	0.808
	Fannie	0.028	0.020	0.025	0.375	0.160	0.410	1.020

Note. The experimental environment consisted of an Intel Core i5-13400F CPU, 32 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU. Training time comparisons were averaged over 10 independent runs, and durations are reported in seconds.

Table 12
Performance comparison among AAESS, AugBoost-RFS, Hybrid-Credit, and DQN-CMDRF.

Dataset	Model	Acc (↑)	AUC (↑)	HM (↑)	Brier score (↓)
Fannie	AAESS	**0.94061**	**0.87963**	0.28890	**0.04705**
	AugBoost-RFS (Liu et al., 2022)	0.92400	0.83190	**0.56030**	0.06140
	Hybrid-Credit (Yang et al., 2024)	0.93231	0.85577	0.42460	0.05423
	DQN-CMDRF (Wang et al., 2022)	0.92451	0.83531	0.43961	0.04737
Give	AAESS	**0.96650**	**0.94484**	**0.67619**	**0.02788**
	AugBoost-RFS	0.93760	0.86880	0.60390	0.04880
	Hybrid-Credit	0.95205	0.90682	0.64005	0.03834
	DQN-CMDRF	0.95363	0.95731	0.63634	0.03974
Bankfear	AAESS	**0.92319**	**0.94725**	**0.82617**	**0.06198**
	AugBoost-RFS	0.91790	0.94480	0.70670	0.06530
	Hybrid-Credit	0.92055	0.94603	0.76644	0.06364
	DQN-CMDRF	0.91976	0.93452	0.77685	0.07463
Shandong	AAESS	**0.96650**	**0.94484**	**0.67619**	**0.02788**
	AugBoost-RFS	0.95230	0.92850	0.65860	0.03350
	Hybrid-Credit	0.96142	0.93143	0.66354	0.03236
	DQN-CMDRF	0.94420	0.92452	0.64840	0.03495

the training and prediction costs of AAESS to a set of benchmark classifiers, as summarized in Table 11. The results were averaged over 10 independent runs. The experimental environment consisted of an Intel Core i5-13400F CPU, 32 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU. Similar to observations in prior work on deep cascade ensembles, the computational cost of credit scoring models was highly dependent on the choice of base learners. Lightweight learners such as DT, LDA, and LR exhibited clear advantages in training and prediction efficiency, with costs remaining at the sub-second level, yet their predictive performance was relatively limited. More advanced ensemble learners such as XGBoost and LightGBM incurred moderately higher computational costs but achieved stronger predictive accuracy. By comparison, AAESS introduced additional runtime overhead due to cross-validation, attention weighting, and auxiliary regression modules. Nevertheless, the training cost of AAESS remained within a practically acceptable range (e.g. 8–76 s across datasets), while the prediction cost (0.17–1.02 s) was still comparable to that of advanced tree-based ensembles. These results indicate that although AAESS is computationally more demanding than simpler baselines, it remains feasible for real-world deployment in credit scoring scenarios where prediction latency is critical, while offering notable improvements in robustness and interpretability.

5.3.3. Performance comparison between the proposed ensemble model and benchmark ensemble models

We then compared the indicators between the baseline ensemble model and other excellent ensemble models. The specific results are shown in Table 12. The AAESS model showcases superior performance in credit scoring tasks by outperforming traditional ensemble methods like random forests, XGBoost, LightGBM, and AdaBoost, as well as advanced models such as SMOTified-AE and Interpretable-AE. It excels across key metrics, particularly at handling imbalanced datasets where default cases are rare, thanks to its strength in cost-sensitive measures like the Brier score and H-measure, alongside robust recall and precision. The model's attention mechanism dynamically prioritizes effective base learners, enhancing adaptability and minimizing both false positives and negatives. This makes AAESS a state-of-the-art solution, effectively balancing accuracy, robustness, and interpretability.

5.4. Statistical comparison of AAESS to other credit scoring models

To validate whether AAESS statistically outperforms existing credit scoring models, we first conducted the Friedman test. The Friedman test is a non-parametric test used for detecting significant differences among models

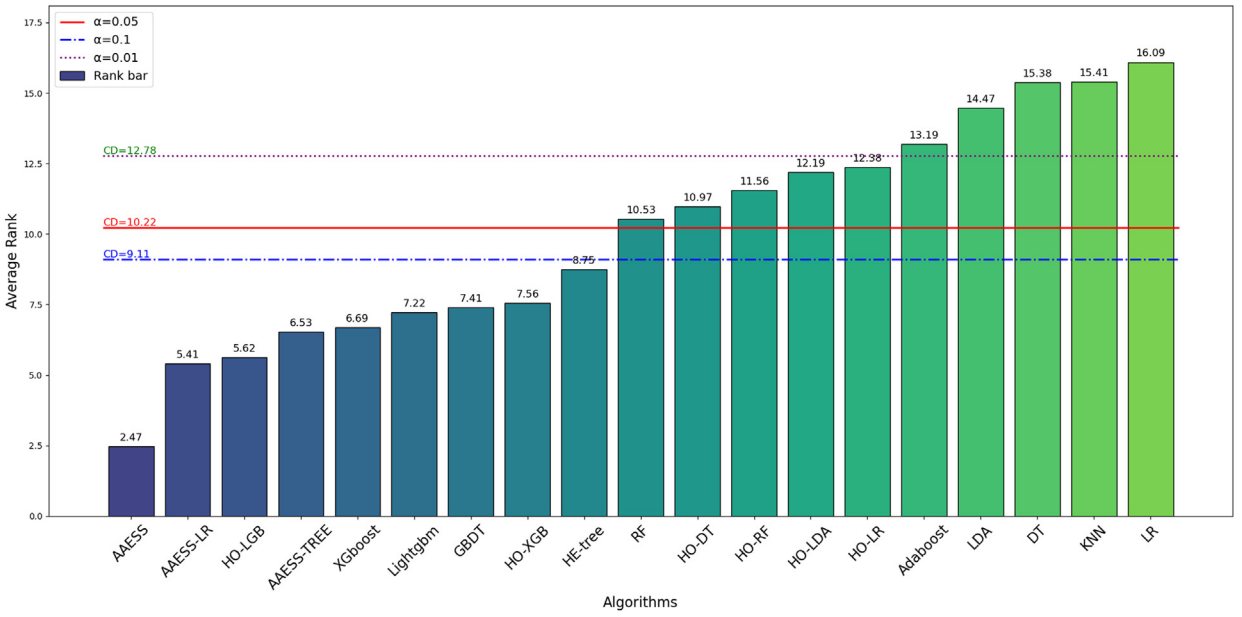


Fig. 6. Average rankings of algorithms and critical differences (CDs) for the Nemenyi test: Horizontal lines indicate CD thresholds and the lowest-ranked algorithm.

by ranking them. It computes the following statistic:

$$\chi_F^2 = \frac{12D}{N_C(N_C+1)}\left[\sum_{N_C=1}^{N_C} AvR_{N_C}^2 - \frac{N_C(N_C+1)^2}{4}\right], \quad (23)$$

where N_C represents the number of classifiers, and D represents the number of credit datasets. The average rank of the N_C-th classifier, AvR_{N_C}, is calculated using the formula

$$AvR_{N_C} = \frac{1}{D}\sum_{d=1}^{D} r_{d,n_C}, \quad (24)$$

where r_{d,n_C} is the rank of the n_C-th classifier on the dth dataset in terms of the four comprehensive credit scoring metrics (i.e. Acc, AUC, BS, and HM). Specifically, r_{d,n_C} can be defined as

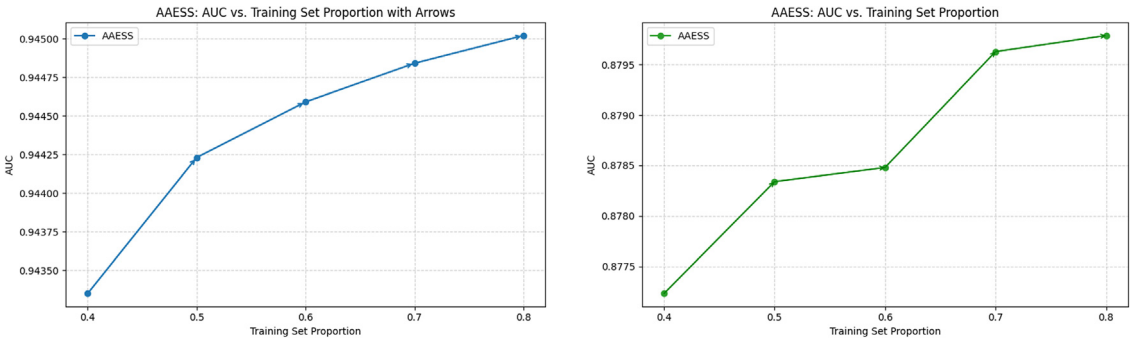
$$r_{d,n_C} = \frac{1}{4}(r_{d,n_C,Acc} + r_{d,n_C,AUC} + r_{d,n_C,BS} + r_{d,n_C,HM}), \quad (25)$$

where $r_{d,n_C,Acc}$, $r_{d,n_C,AUC}$, $r_{d,n_C,BS}$, and $r_{d,n_C,HM}$ denote the rankings of the n_C-th classifier in terms of the Acc, AUC, BS, and HM on the dth dataset, respectively. If the Friedman test statistic is less than the critical threshold, we accept the null hypothesis that no significant differences exist among the credit scoring methods. Otherwise, we reject the null hypothesis and conduct post hoc tests, such as the Nemenyi test, to identify significant differences between AAESS and the baseline models. When the average rankings differ by at least the critical difference (CD), there is a statistically significant difference between the baseline models and AAESS. The critical difference is computed using

$$CD_\alpha = q_\alpha\sqrt{\frac{N_C(N_C+1)}{12D}}, \quad (26)$$

where q_α is the critical value for the chosen confidence level. To validate the proposed AAESS model's superior performance in various quantitative measures compared to other state-of-the-art credit scoring models and advanced ensemble techniques such as LightGBM and XGBoost, we conducted a Friedman test to determine if there were statistically significant differences among these credit scoring algorithms. To evaluate the rankings across different metrics, we first calculated the average rankings based on the Acc, AUC, BS, and HM. The Friedman test statistics calculated were as follows: $\chi_{Acc} = 42.0237$, $\chi_{AUC} = 56.7829$, $\chi_{BS} = 39.6513$, and $\chi_{HM} = 41.2145$. All metrics exceeded the critical value at a 99% confidence level (rejecting the null hypothesis that no significant differences exist among the models). Next, a post hoc procedure (the Nemenyi test) was conducted to identify the specific comparisons. The critical value thresholds for different confidence levels were $q_{0.01} = 3.213$, $q_{0.05} = 2.569$, and $q_{0.1} = 2.291$, respectively. For $\alpha = 0.01$, $\alpha = 0.05$, and $\alpha = 0.1$, the critical differences (CD) were calculated as follows: $CD_{0.01} = 12.78$, $CD_{0.05} = 10.22$, and $CD_{0.1} = 9.11$.

Fig. 6 illustrates the average rankings and CDs for various credit scoring methods calculated using the Nemenyi test at different significance levels. As shown in the figure, AAESS consistently outperforms other models, including those dominated by LightGBM and XGBoost, achieving the best overall performance in credit scoring. The average rank of AAESS is 2.47, while AAESS-LR holds the second-best average rank, providing further evidence of the potential of AAESS as a strong framework for credit scoring. At the significance level of $\alpha = 0.01$, AAESS outperforms Adaboost, LDA, DT, KNN, LR, and other models. At the significance level of $\alpha = 0.05$, AAESS outperforms models like RF, Adaboost, DT, KNN, LR, and other



(a) Give (b) Shandong

Fig. 7. Sensitivity comparison across datasets.

homogeneous ensemble models (HO-DT, HO-RF, HO-LDA, and HO-LR). The ranking of the heterogeneous models also surpasses that of the homogeneous models, further proving the potential of AAESS. This demonstrates that compared to homogeneous models, heterogeneous structures are a highly robust option. The strengths of combining different types of base classifiers show that AAESS's ranking verifies its potential as a leading model for credit scoring.

5.5. Sensitivity analysis

Benefiting from the adaptive attention-enhanced sequential stacking structure and robust feature optimization via Pareto-based feature selection, AAESS is designed as a flexible stacking ensemble whose predictive capability scales effectively with the size of available data. These features highlight AAESS as a compelling method for modeling credit risks, particularly in scenarios involving limited data. To investigate the scalability and robustness of AAESS on small-scale credit datasets, we analyzed its performance across different training set proportions. Fig. 7(a) and 7(b) present the performance of AAESS across varying training data ratios on the two credit datasets: Shandong and Give. It can be seen from Fig. 7(a) that even with as little as 40% of the training data, AAESS maintains consistently high AUC values across all datasets. Specifically, on the Give dataset, AAESS achieves an impressive AUC value exceeding 0.94 with only 40% of the training data. Furthermore, increasing the training set size from 40% to 80% yields only marginal improvements, demonstrating a minimal decay in predictive performance and underscoring the robustness and adaptability of AAESS in small-data contexts. Finally, the consistently tight AUC band highlights the stability of AAESS, a property that translates into smoother regulatory backtesting cycles and lower model-risk capital charges.

5.6. Validation on a small-sample credit dataset

The preceding analysis demonstrated that AAESS remains robust when the effective training size of large-scale datasets is reduced. To validate its applicability, we extend our evaluation to an inherently small-sample benchmark. Such scenarios are highly relevant in non-retail credit markets such as small and medium-sized enterprises or commercial lending, where historical data availability is limited and model generalization becomes particularly challenging.

Table 13
Summary of Australian Credit Approval dataset.

Dataset	Samples	Features	Good/bad
Australian	690	14	307/383

Table 14
Performance metrics for different algorithms on the Australian Credit Approval dataset.

Model	Acc ↑	AUC ↑	BS ↓	F1-score ↑	Type-II ↓
LR	0.8551	0.9096	0.1581	0.8274	0.1278
LDA	0.8565	0.9171	0.2022	0.8750	0.1153
DT	0.8348	0.8252	0.1985	0.8447	0.1556
KNN	0.8420	0.9280	0.1349	0.8533	0.1111
LightGBM	0.8754	0.9360	0.1252	0.8947	0.1050
XGBoost	0.8652	0.8983	0.1153	0.8056	0.1153
GBDT	0.8696	0.8996	0.1084	0.8042	0.1084
AdaBoost	0.8536	0.8846	0.1625	0.8010	0.1850
RF	0.8710	0.8839	0.1075	0.8030	0.1075
AAESS	**0.8841**	**0.9484**	**0.1003**	**0.8919**	**0.0833**

To this end, we conducted a comprehensive comparison on the Australian Credit Approval dataset (see Table 13), a widely used benchmark for small-sample credit scoring (http://archive.ics.uci.edu/ml). The experimental results, summarized in Table 14, provide a direct assessment of the model's ability to cope with data scarcity.

As shown in Table 14, AAESS consistently outperforms the competing models, achieving the highest AUC and lowest Brier score. These results highlight its superior ranking capability and probability calibration, both of which are crucial for accurate risk pricing and capital allocation in practice. Although tree-based ensembles such as LightGBM and XGBoost demonstrate competitive accuracy, their performance is more susceptible to overfitting in data-scarce environments. In contrast, the adaptive weighting and Pareto-optimal feature selection embedded in AAESS confer enhanced generalization and robustness.

Table 15
Performance and interpretability analysis of AAESS on the Give dataset.

Metric	Value
Accuracy (Acc)	0.9412
AUC	0.8811
LogLoss	0.2469
KS	0.5493
Precision	0.4786
Recall	0.2278
F1-score	0.3072
Brier score	0.0468
Average precision	0.3514
Top meta-coefficients (|value|)	Coefficient
RF_awp	44.216
LGB_awp	3.485
XGB_awp	3.294
Average attention (test)	Value
XGB	0.3375
RF	0.3302
LGB	0.3323
Normalized learner contributions	Value
XGB	7.21%
RF	84.96%
LGB	7.83%

This experiment strengthens the empirical evidence that AAESS is not merely a mechanical ensemble of machine learning techniques, but a framework deliberately tailored for the realities of credit risk modeling. Its consistent performance across both large-scale retail datasets and inherently small-sample credit datasets underscores its versatility and domain-specific value.

5.7. Interpretability of AAESS

Credit assessment models are critical tools that evaluate an applicant's ability to repay financial debts, enabling lending institutions to make informed loan decisions (Chen et al., 2024; Li et al., 2022; Xia et al., 2025). Advanced machine learning-based credit scoring models have improved the efficiency of loan approvals by targeting specific applicant segments, reducing costs, and minimizing human subjectivity in default risk evaluation. However, these models are frequently criticized as black boxes, due to their lack of transparency, which hinders trust among financial experts and limits practical adoption. The AAESS model overcomes this limitation by prioritizing interpretability within its framework. Its transparent Pareto-optimal feature selection process explicitly identifies and justifies the most predictive features, balancing multiple objectives such as accuracy and robustness, while allowing users to understand the rationale behind each feature's inclusion. Additionally, the attention mechanism dynamically weights the contributions of base learners, providing clear visibility into how individual predictions are aggregated and illuminating the decision-making process. By integrating Bayesian and Huber regression, AAESS further enhances reliability by mitigating the effects of noise and outliers, with sample and model-level weights offering traceable insights into data reliability and learner performance. This interpretability not only fosters trust among practitioners but also aligns with regulatory requirements for explainable AI in finance, making AAESS a highly practical and reliable solution for credit scoring applications.

5.8. Interpretability of attention

While the incorporation of attention-weighted base learner predictions inevitably increases the model's complexity, we emphasize that interpretability is not lost. In particular, the meta-classifier remains a logistic regression model, which provides transparent coefficients that quantify the contribution of both original features and the attention-weighted predictions. For the interpretability analysis, we report only the three major base learners. Although the AAESS framework employs four base learners during training and inference, the contribution of AdaBoost is less than 1% after normalization. Such a minimal influence has a negligible effect on the meta-classifier's decision process, and including it in the interpretability tables would not change any conclusion while reducing readability. Therefore, we focus on the three dominant learners, while the full four-learner configuration is still used during prediction.

To empirically validate this transparency, we conducted a detailed interpretability analysis on the GiveMeSomeCredit dataset. The results, summarized in Table 15, demonstrate that the model achieves strong predictive performance (Acc = 0.9412, AUC = 0.8811, LogLoss = 0.2469, KS = 0.5493) while retaining explicit explainability. The logistic regression coefficients are dominated by the attention-weighted outputs of the base learners (e.g. $RF_{awp} = 44.22$, $LGB_{awp} = 3.49$, $XGB_{awp} = 3.29$), indicating how the meta-classifier prioritizes aggregated predictions. Furthermore, the average attention weights remain nearly uniform across the three base learners ($XGB = 0.3375$, $RF = 0.3302$, $LGB = 0.3323$), demonstrating a balanced initial treatment at the attention layer. However, when normalized by the logistic regression coefficients, the effective learner contributions reveal that RF accounts for 84.96% of the final prediction influence, with XGB and LGB contributing 7.21% and 7.83%, respectively. These findings illustrate that the AAESS framework maintains regression-style interpretability while leveraging attention to enhance robustness.

We extended this detailed interpretability analysis to the remaining three datasets—Shandong, Fannie Mae, and Bankfear—to verify the generalizability of these findings, as shown in Tables 16, 17, and 18. Across these diverse financial environments, a consistent structural pattern emerged: the random forest learner invariably dominated the final decision weights, accounting for approximately 82% to 89% of the normalized learner contribution, despite the attention mechanism maintaining a relatively balanced distribution of weights among all base learners.

We conjecture that this robust phenomenon stems from the structural diversity embedded within the ensemble. Specifically, XGBoost and LightGBM, both belonging to the gradient boosting family, aim to reduce bias and likely generate highly correlated prediction manifolds;

Table 16
Performance and interpretability analysis of AAESS on the Shandong dataset.

Metric	Value
Accuracy (Acc)	0.9665
AUC	0.9448
LogLoss	0.1256
KS	0.7642
Precision	0.6580
Recall	0.5708
F1-score	0.6113
Brier score	0.0279
Average precision	0.5924
Top meta-coefficients (|value|)	Coefficient
RF_awp	4.6121
LGB_awp	3.0440
XGB_awp	2.4300
Average attention (test)	Value
XGB	0.3351
RF	0.3310
LGB	0.3339
Normalized learner contributions	Value
XGB	6.12%
RF	86.58%
LGB	7.30%

Table 17
Performance and interpretability analysis of AAESS on the Fannie Mae dataset.

Metric	Value
Accuracy (Acc)	0.9249
AUC	0.8445
LogLoss	0.3105
KS	0.4821
Precision	0.5120
Recall	0.0629
F1-score	0.1121
Brier score	0.0611
Average precision	0.3845
Top meta-coefficients (|value|)	Coefficient
RF_awp	39.540
LGB_awp	4.120
XGB_awp	3.855
Average attention (test)	Value
XGB	0.3362
RF	0.3298
LGB	0.3340
Normalized learner contributions	Value
XGB	8.58%
RF	82.15%
LGB	9.27%

Table 18
Performance and interpretability analysis of AAESS on the BankFear dataset.

Metric	Value
Accuracy (Acc)	0.9232
AUC	0.9473
LogLoss	0.1985
KS	0.7812
Precision	0.7854
Recall	0.7696
F1-score	0.7774
Brier score	0.0620
Average precision	0.8105
Top meta-coefficients (|value|)	Coefficient
RF_awp	52.330
LGB_awp	2.850
XGB_awp	2.645
Average attention (test)	Value
XGB	0.3380
RF	0.3285
LGB	0.3335
Normalized learner contributions	Value
XGB	5.10%
RF	89.24%
LGB	5.66%

this multicollinearity causes the linear meta-classifier to distribute or suppress their individual coefficients. In contrast, random forests, as a bagging technique focused on variance reduction, provides a distinct and stable probabilistic baseline. Consequently, the AAESS framework effectively utilizes random forests as a robust 'anchor' for the decision boundary, while leveraging the boosting models to provide fine-grained corrections for complex instances. This confirms that the model successfully integrates the complementary strengths of bagging (stability) and boosting (sensitivity), rather than merely averaging redundant learners.

6. Conclusion and future work

By leveraging known historical information about borrowers, it is possible to accurately assess their repayment capacity, effectively preventing financial losses for banks and lending institutions due to insufficient repayment ability. This study aimed to integrate the attention mechanism from deep learning with statistical predictive models to avoid prediction bias caused by insufficient features while improving model accuracy. We first proposed a Pareto-optimal feature selection method that comprehensively considers multiple evaluation metrics to enhance model versatility. Additionally, we introduced the adaptive attention-enhanced sequential stacking (AAESS) model, which combines the attention mechanism with advanced feature selection techniques to improve credit scoring effectiveness.

For variable selection, the model employs a Pareto-optimal framework and five feature selection methods, namely classifier-based feature evaluation, correlation-based feature weighting, gain ratio feature evaluation, information gain feature evaluation, and relief feature evaluation. By integrating these five feature selection methods and using random forests to assess feature importance, we were able to filter out redundant or irrelevant features, refine the dataset, and enhance the predictive capability of the model.

To ensure model stability, we employed four methods to detect outliers in the dataset. For significant outliers, the Huber model was used to perform adaptive adjustments, while within the normal feature range, a Bayesian model was applied for adaptive weight distribution. In the stacking model, random forests, XGBoost, LightGBM, and Adaboost were selected as base learners. The research

findings indicate that the stacked ensemble model can predict credit default risk more accurately than individual predictive models. By comparing the AAESS model with single classifiers and traditional forecast combinations, we found that AAESS was statistically superior to other models.

Finally, we performed a statistical analysis of the adaptive decision weights, highlighting the distinct contribution of each base learner to the final prediction. This quantitative weight analysis helps users intuitively understand the model's internal decision dynamics and its reliance on robust estimators. The AAESS model demonstrated excellent capability at balancing multiple performance metrics, making it an effective solution for modern credit scoring applications. The results of this study provide valuable insights into the potential of combining attention mechanisms and ensemble learning for optimizing credit scoring models, with broad applications in financial risk assessment and management.

Although the AAESS model improves credit scoring performance, there are three key areas for future enhancement:

(1) Improving model interpretability: Despite the AAESS model's excellent predictive accuracy and performance, its complexity may limit user understanding of the decision-making process. Future work can focus on incorporating more advanced interpretability techniques, such as visualization tools or locally interpretable models, to help users better understand the contribution of features to the prediction results, thereby enhancing the model's operability and trustworthiness in practical applications.

(2) Enhancing external validity and robustness to concept drift: Despite the comprehensive empirical evaluation across heterogeneous credit environments in this study, we candidly acknowledge that the external validity of the AAESS framework is intrinsically bounded by the empirical data distributions it is trained on. As a data-driven predictive model, it remains vulnerable to sudden macroeconomic regime shifts or rapid concept drift, such as abrupt changes in institutional lending policies. Consequently, the predictive relationships and static base-learner weights learned during training may degrade over time, necessitating periodic recalibration when deployed in real-world forecasting systems. Future research should explore the integration of adaptive concept-drift detection mechanisms or online learning strategies to enhance the external validity and long-term robustness of the ensemble in dynamic financial environments.

(3) Optimizing computational efficiency and real-time application capability: The AAESS model involves multilayer stacking and feature selection, which may result in higher computational costs in real-time applications. Future work could explore optimizing the algorithm's computational efficiency while maintaining predictive performance, reducing model training and inference time to better suit real-time financial risk assessments and credit scoring systems, meeting the need for rapid decision making.

These improvements would contribute to enhancing the practicality and potential for widespread adoption of the AAESS model.

CRediT authorship contribution statement

Lingyi Meng: Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Shanshan Jiang:** Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Min Xia:** Writing – review & editing, Software, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Outlier detection strategy

The interquartile range (IQR) method is used to detect outliers by calculating the distance between the first quartile (Q1) and the third quartile (Q3) in a dataset. Outliers are defined by the lower and upper bounds:

$$\mathrm{IQR} = Q3 - Q1, \tag{27}$$

$$\text{Lower Bound} = Q1 - 1.5 \times \mathrm{IQR}, \tag{28}$$

$$\text{Upper Bound} = Q3 + 1.5 \times \mathrm{IQR}. \tag{29}$$

Data points outside this range are considered outliers. Another method, the Z-score, measures how far each data point is from the mean in terms of standard deviations:

$$Z = \frac{X - \mu}{\sigma}, \tag{30}$$

where μ is the mean and σ is the standard deviation. Values with a Z-score greater than a set threshold are flagged as outliers. Additionally, the isolation forest method, which partitions data to isolate outliers, and the local outlier factor (LOF) method, which detects outliers based on local density, further enhance the outlier detection process, especially in high-dimensional or complex datasets.

To integrate these four detectors into a unified decision, we employ a majority-voting scheme: each method produces a binary flag (outlier vs. inlier), and a sample is ultimately labeled an outlier if it is identified by at least three of the four methods. This ensemble aggregation reduces the sensitivity to the biases of individual methods and improves robustness. After aggregation, the overall outlier ratio is computed, and if this ratio exceeds 5%, Huber regression is selected; otherwise, Bayesian ridge regression is employed for adaptive sample weighting, as illustrated in Fig. 1.

The adoption of a 5% threshold for outlier handling is motivated by both statistical theory and established empirical practice. In classical statistics, observations beyond approximately the 95th percentile (or below the 5th percentile) are conventionally regarded as outliers, which corresponds to the well-known $\pm 1.96\sigma$ rule that flags roughly 5% of extreme values under a normal distribution (Shao et al., 2018). Robust estimation methods also embody this principle. For example, Huber's M-estimator

Table A.19
Ablation on outlier handling: AAESS vs. AAESS-ZscoreFilter (Z-score filtering with removal). Metrics on four datasets. The best numbers per dataset are in bold.

Dataset	Method	Accuracy ↑	AUC ↑	BS ↓	HM ↑	Type-I ↓	Type-II ↓
Bankfear	AAESS	**0.92319**	**0.94725**	**0.06198**	**0.82617**	**0.02905**	**0.23039**
	AAESS-ZscoreFilter	0.91589	0.92917	0.07441	0.80998	0.03260	0.27043
Fannie	AAESS	**0.92493**	**0.84452**	**0.06107**	**0.09226**	**0.00067**	**0.93708**
	AAESS-ZscoreFilter	0.92254	0.82715	0.06284	0.08036	0.00288	0.95672
Give	AAESS	**0.94061**	**0.87963**	**0.04705**	**0.28890**	**0.00607**	**0.78571**
	AAESS-ZscoreFilter	0.93553	0.86830	0.04986	0.21061	0.00664	0.87139
Shandong	AAESS	**0.96650**	**0.94484**	**0.02788**	**0.67619**	**0.00294**	**0.42923**
	AAESS-ZscoreFilter	0.95150	0.91317	0.03930	0.49741	0.00403	0.65091

Note. AAESS-ZscoreFilter replaces the robust outlier-weighting module with simple Z-score filtering (threshold $|z| > 3$) and removes flagged samples. The numbers for AAESS are extracted from Tables 6–9, while the Z-score results are obtained from our additional ablation experiments. The results show that AAESS consistently outperforms Z-score filtering across all datasets, demonstrating the effectiveness of its robust outlier handling strategy.

is typically tuned to achieve approximately 95% efficiency under normality, effectively down-weighting the most extreme 5% of observations while retaining the efficiency of ordinary least squares on uncontaminated data. This design reflects a compromise: it safeguards against small-scale contamination while preserving information from the majority of the sample (Raza et al., 2024).

From an applied perspective, trimming or winsorizing the top and bottom 5% of a distribution is a widely used data-cleaning strategy in econometrics and finance. Such practices are recommended to mitigate the disproportionate influence of extreme values while maintaining approximately 90% of the central data for analysis. In fact, guidelines often advise that removal beyond 5% per tail may indicate deeper issues in data collection rather than routine noise, underscoring the role of 5% as a pragmatic upper bound for outlier exclusion.

The prevalence of this convention is further supported by the literature. Simulation studies of robust regression methods frequently adopt contamination levels of 5% and 10% to evaluate resilience, reinforcing the interpretation of 5% as a representative moderate contamination level. Empirical research across diverse domains, including finance and credit risk modeling, similarly applies the 5% rule to exclude or adjust extreme observations and thereby stabilize estimation (Ayyagari et al., 2018).

Overall, the choice of a 5% threshold provides a transparent, literature-backed rationale for outlier-driven method switching. It is sufficiently large to capture aberrant points that could disproportionately distort regression outcomes, yet sufficiently conservative to avoid excessive data loss. Within our framework, this threshold governs the transition between Bayesian ridge regression and Huber regression, ensuring that sample weights and model confidence are computed under assumptions aligned with the contamination level of the data. This design strengthens both the robustness and interpretability of the final predictions.

A.1. Ablation study on outlier handling

To validate the contribution of the proposed robust outlier detection and weighting mechanism, we design an ablation study that compares AAESS against a simplified baseline method. Specifically, the baseline adopts a conventional Z-score filtering strategy, in which any sample with a feature value exceeding three standard deviations from the mean is regarded as an outlier and removed from the training set. This variant, denoted as AAESS-ZscoreFilter, maintains the overall AAESS architecture but replaces the robust weighting module with simple Z-score-based exclusion.

For the comparison, the full AAESS model employs an ensemble of four detectors (IQR, Z-score, isolation forest, and LOF) combined via majority voting, with suspected outliers adaptively down-weighted through Huber or Bayesian ridge regression rather than being discarded. Both model variants are evaluated across all benchmark datasets using the AUC, Brier score, H-measure, and accuracy as performance criteria.

This design allows us to isolate the marginal effect of the proposed outlier handling strategy. We hypothesize that AAESS will consistently outperform AAESS-ZscoreFilter, since the outright removal of data points risks discarding informative but extreme samples, whereas adaptive down-weighting reduces their influence while retaining valuable information for model training. The results of this ablation study are reported in Table A.19, which provides quantitative evidence of the effectiveness of the proposed module.

Data and code availability

The data and code necessary to reproduce the results gathered in this manuscript are available at https://github.com/SHIMMER-0039/credit-scoring.git.

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